Unit

22 Electrolysis

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Factors affecting the order of discharge of ions
- Industrial uses of electrolysis
- Environmental impact of the electroplating industry

22.1 Electrolysis: chemical reactions from electricity

We have learnt that chemical reactions can be used to generate electricity. In this unit we will learn how to use electricity to bring out a chemical reaction. The process is called **electrolysis**.

Electrolysis is an endothermic process. The energy source is electrical energy rather than heat.

The following terms are commonly used in electrolysis.

- **Electrolysis** is the chemical reaction that occurs when electricity passes through an electrolyte in molten state or in aqueous solution.
- An **electrolytic cell** is a cell in which electrolysis occurs.
- An **electrolyte** is a substance which conducts electricity in molten state or in aqueous solution, and is decomposed by electricity during the conduction.
- The **anode** is the electrode where oxidation occurs. The anode of an electrolytic cell is the electrode connected to the positive electrode of the d.c. supply (Fig. 22.1). Electrons are withdrawn from the anode by the power source. Anions will discharge (undergo oxidation) here.
- The **cathode** is the electrode where reduction occurs. The cathode of an electrolytic cell is the electrode connected to the negative electrode of the d.c. supply (Fig. 22.1). Electrons flow from the power source to the cathode. Cations will discharge (undergo reduction) here.

electrolysis 電解　electrolytic cell 電解池　electrolyte 電解質　anode 陽極　cathode 陰極

The symbol Ⓐ in the circuit represents an **ammeter**, an instrument used to measure the electric current passing through a circuit. The symbol ⏥ represents a **variable resistor**, an instrument used to vary the resistance of the circuit and hence regulate the current.

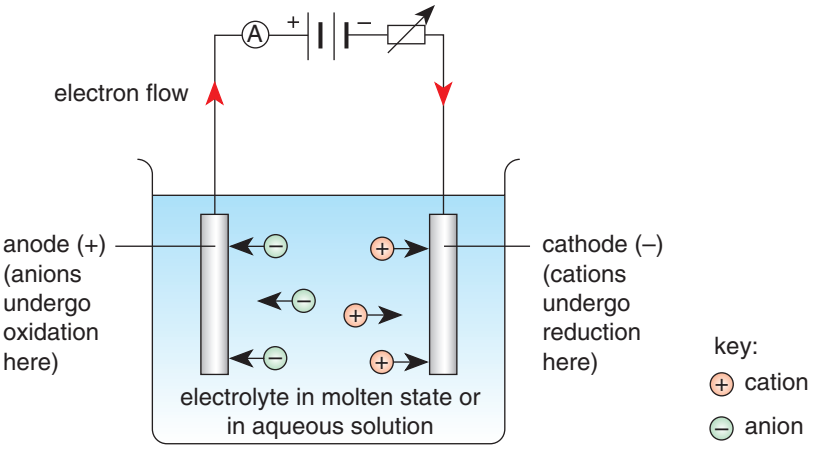


Fig. 22.1 An electrolytic cell

22.2 Comparing a chemical cell and an electrolytic cell

A chemical cell and an electrolytic cell have some common characteristics and some differences (Table 22.1).

A chemical cell converts chemical energy into electrical energy through chemical changes that occur at the electrodes. In contrast, an electrolytic cell requires an external energy source to operate.

Table 22.1

A comparison between a chemical cell and an electrolytic cell N1

Chemical cell	Electrolytic cell
produces electrical energy	requires electrical energy to drive the reaction
oxidation occurs at the anode, the *negative* electrode	oxidation occurs at the anode, the *positive* electrode
reduction occurs at the cathode, the *positive* electrode	reduction occurs at the cathode, the *negative* electrode

CE 2011 Paper 1 Q10(b)(i)

N1 When using a zinc-carbon cell in electrolysis, the zinc electrode of the cell is connected to the cathode of the electrolytic cell.

ammeter 安培計　　variable resistor 變阻器

Electrodes used in electrolysis are usually made of inert substances such as carbon or platinum. Inert electrodes do not react with the electrolyte or the products of electrolysis.

22.3 Electrolysis of molten sodium chloride using carbon electrodes

Solid sodium chloride does not conduct electricity. However, when electricity is passed through molten sodium chloride, a chemical reaction occurs. A shiny bead of sodium is produced at the cathode and chlorine gas is evolved at the anode. This process is an example of electrolysis.

During the electrolysis, the chloride ions are attracted to the anode where they undergo oxidation. The sodium ions are attracted to the cathode where they undergo reduction.

At the anode:

$2Cl^-(l) \longrightarrow Cl_2(g) + 2e^-$ oxidation

At the cathode:

$Na^+(l) + e^- \longrightarrow Na(l)$ N2 reduction

N2 Although students are NOT required to perform calculations on electrolysis, examination questions may ask students to deduce the oxidation number of an element based on the amount of products formed during electrolysis.

For example, consider the hydride of element Z. During the electrolysis of the molten compound, the following process occurs at the anode:

$2H^-(l) \longrightarrow H_2(g) + 2e^-$

It is known that the number of Z atom formed at the cathode equals the number of hydrogen molecules formed at the anode (i.e. number of moles of Z atom formed = number of moles of H_2 molecules formed).

Suppose 0.1 mole of hydrogen molecules is formed at the anode, it can be deduced from the above equation that 0.2 mole of electrons are involved. Then it can be deduced that 0.2 mole of electrons are involved in the production of 0.1 mole of Z atoms. The following process occurs at the cathode:

$Z^{2+}(l) + 2e^- \longrightarrow Z(s)$

Hence it can be deduced that the oxidation number of Z in the hydride is +2.

22.4 Some knowledge related to aqueous electrolytes

Dissociation of water

Water dissociates slightly to give hydrogen ions and hydroxide ions.

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

Dissociation of acids in water

Acids are covalent compounds. However, acid molecules dissociate in water to give ions.

We have discussed this in Topic 4 Acids and Bases.

For example,

$$\underset{\text{molecule of hydrogen chloride}}{HCl(g)} + \text{water} \longrightarrow \underbrace{H^+(aq) + Cl^-(aq)}_{\text{mobile ions}}$$

Dissolving electrolytes in water

Mobile ions are formed when we dissolve electrolytes in water. For example, solid sodium chloride gives mobile sodium ions and chloride ions in water.

$$NaCl(s) + water \longrightarrow \underbrace{Na^+(aq) + Cl^-(aq)}_{\text{mobile ions}}$$

22.5 Electrolysis of aqueous solutions of ionic compounds

When a molten salt is electrolyzed, the products are predictable. However, during the electrolysis of an aqueous solution of an ionic compound, ions from water may *discharge preferentially* at the electrodes.

Electrolysis of acidified water using platinum electrodes

Fig. 22.2 shows the electrolysis of acidified water using a *Hoffman voltameter*.

22.1
Investigating the electrolysis of acidified water.

Video
Electrolysis of acidified water

Reference Website
Refer to the following website for an animation illustrating the chemical reactions that occur during the electrolysis of acidified water:
http://my.rsc.org/video/288
(accessed July 2014)

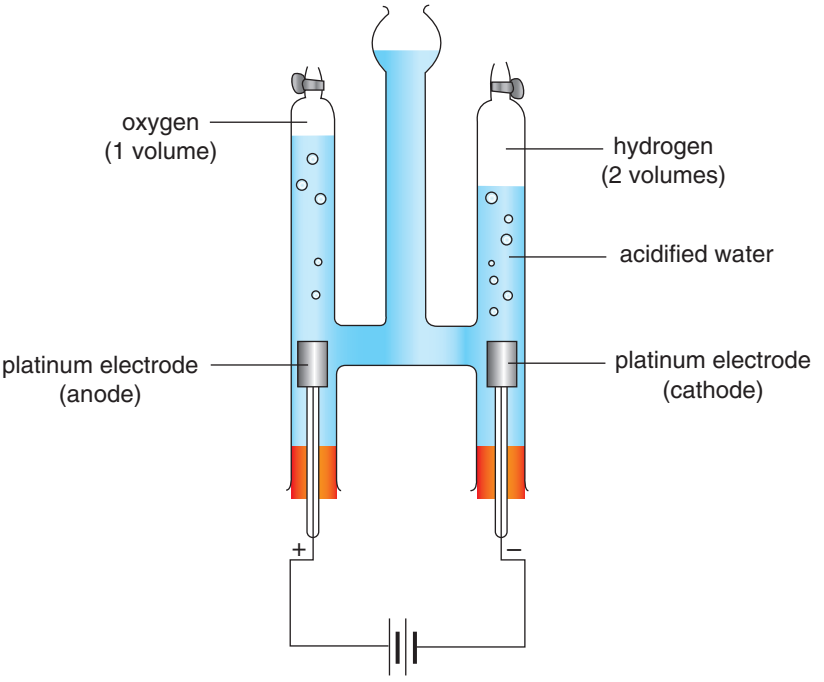


Fig. 22.2 Electrolysis of acidified water using a Hoffman voltameter

preferential discharge 優先放電　　Hoffman voltameter 霍夫曼電量計

Pure water is a very poor conductor of electricity because it only dissociates slightly. When we add some dilute sulphuric acid to pure water, the resulting solution becomes a good conductor of electricity because the amount of mobile ions increases.

There are three kinds of ions in acidified water: $H^+(aq)$, $OH^-(aq)$ and $SO_4^{2-}(aq)$.

Ions attracted to the anode	$OH^-(aq)$, $SO_4^{2-}(aq)$
Ions attracted to the cathode	$H^+(aq)$

At the anode

Hydroxide ions are preferentially discharged (oxidized) to form oxygen gas.

$$4OH^-(aq) \longrightarrow O_2(g) + 2H_2O(l) + 4e^-$$

At the cathode

Hydrogen ions are discharged (reduced) to form hydrogen gas.

$$2H^+(aq) + 2e^- \longrightarrow H_2(g)$$

Overall cell reaction

$$2H_2O(l) \longrightarrow 2H_2(g) + O_2(g)$$

According to the equation, the number of moles of hydrogen : number of moles of oxygen is 2:1. Notice that volume of hydrogen : volume of oxygen is also 2:1.

You will understand why this volume ratio is the same as the H : O mole ratio when studying Topic 10 Rate of Reaction.

Changes in the solution

When hydrogen ions and hydroxide ions are discharged, more water molecules dissociate to replace the hydrogen ions and hydroxide ions discharged. The net effect is that water is decomposed.

The numbers of hydrogen ions and sulphate ions from the sulphuric acid remain the same. The concentration of sulphuric acid increases at the end as water is consumed in the electrolysis.

Electrolysis of very dilute sodium chloride solution using carbon electrodes N3

Fig. 22.3 shows an electrolytic cell for the electrolysis of very dilute sodium chloride solution.

The concentration of the sodium chloride solution used is less than 0.1 mol dm^{-3}.

Animation
Electrolysis of very dilute sodium chloride solution using carbon electrodes

N3 Carbon electrodes used in electrolysis
Carbon is commonly used to make electrodes used in electrolysis because
- it is a good conductor of electricity;
- it does not form ions and is not attacked by chlorine;
- it can be used in the electrolysis of hot molten compounds because of its very high melting point;
- it is cheap.

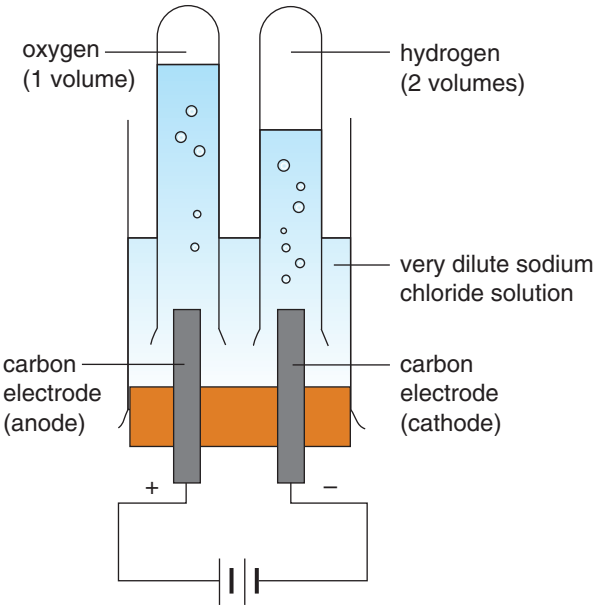


Fig. 22.3 Electrolysis of very dilute sodium chloride solution

In very dilute sodium chloride solution, there are four kinds of ions: $Na^+(aq)$, $Cl^-(aq)$, $H^+(aq)$ and $OH^-(aq)$.

Ions attracted to the anode	$Cl^-(aq)$, $OH^-(aq)$
Ions attracted to the cathode	$Na^+(aq)$, $H^+(aq)$

At the anode

CE 2005 Paper 1 Q9(b)(ii)

Hydroxide ions are preferentially discharged (oxidized) to form oxygen gas.

$$4OH^-(aq) \longrightarrow O_2(g) + 2H_2O(l) + 4e^-$$

We will discuss the order of discharge of ions in Section 22.7.

At the cathode

Hydrogen ions are preferentially discharged (reduced) to form hydrogen gas.

$$2H^+(aq) + 2e^- \longrightarrow H_2(g)$$

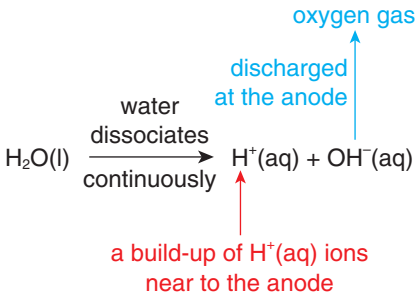


Fig. 22.4a A build-up of $H^+(aq)$ ions near to the anode

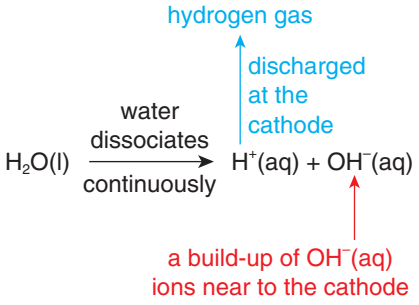


Fig. 22.4b A build-up of $OH^-(aq)$ ions near to the cathode

Overall cell reaction

$$2H_2O(l) \longrightarrow 2H_2(g) + O_2(g)$$ CE 2008 Paper 2 Q48

Changes in the solution

Water dissociates continuously to replace the hydroxide ions discharged at the anode.

Thus, there is a build-up of hydrogen ions near to the anode (Fig. 22.4a) and the solution there becomes acidic.

At the same time, water dissociates continuously to replace the hydrogen ions discharged at the cathode. Thus, there is a build-up of hydroxide ions near to the cathode (Fig. 22.4b). The solution there becomes alkaline.

If we add a few drops of universal indicator solution to the sodium chloride solution, the solution near to the anode will turn red while the solution near to the cathode will turn blue.

The sodium chloride solution becomes more concentrated as water is decomposed in the electrolysis.

Practice 22.1

P22.1 Refer to Teacher's Resources Book page 56.

An experiment was carried out to study the electrolysis of dilute potassium sulphate solution. Dilute potassium sulphate solution with some universal indicator solution was electrolyzed using the set-up shown. N4

a) Hydrogen gas was formed at electrode X.

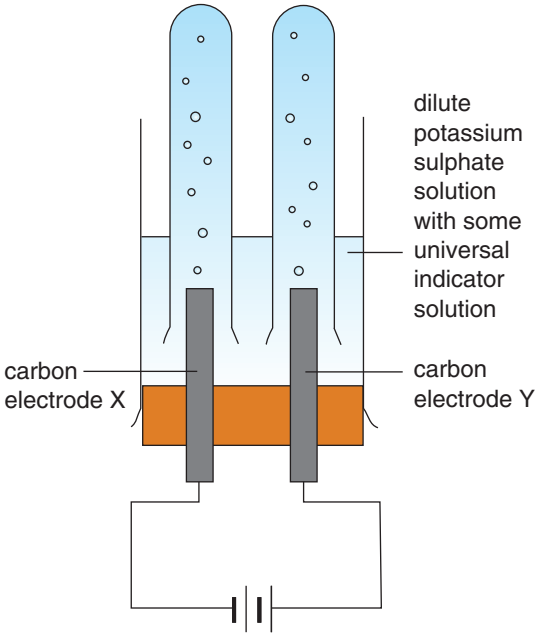
   i) Write an ionic half-equation for the reaction that occurred at electrode X.

   ii) Explain why the solution near to electrode X turned blue after some time.

b) Oxygen gas was formed at electrode Y.

   i) Write an ionic half-equation for the reaction that occurred at electrode Y.

   ii) Explain why the solution near to electrode Y turned red after some time.



N4 The electrolysis of a saturated solution of potassium hydrogen sulphate ($KHSO_4$) produces potassium peroxodisulphate ($K_2S_2O_8$) at the anode.

$2KHSO_4 \longrightarrow \underset{\text{cathode}}{H_2} + \underset{\text{anode}}{K_2S_2O_8}$o

N5 Rate of electrolysis

Electrolysis, in general, can be speeded up by increasing the electric current flowing through the circuit. Some possible methods are:

1 to increase the voltage applied for electrolysis.
2 to decrease the total resistance of the circuit. This can be done by:
- moving the electrodes closer together;
- pushing the electrodes deeper into solution so that more surface area will be in contact with the solution;
- using electrodes of larger surface area;
- decreasing the resistance in the rheostat, if any;
- increasing the concentration of the electrolyte.

22.6 Factors affecting the order of discharge of ions during the electrolysis of aqueous solutions N5

Which ions are preferentially discharged during the electrolysis of aqueous solutions depends on a number of factors, including:

- the position of ions in the electrochemical series;
- the concentration of ions in the solution; and
- the nature of the electrodes used.

Alteration of any of these factors may change the order of discharge of ions and thus the nature of the products.

22.7 The position of ions in the electrochemical series and the order of discharge of ions

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

22.2

Investigating factors affecting the order of discharge of ions during electrolysis — position of ions in the electrochemical series.

Order of discharge of cations

Fig. 22.5 shows the order of discharge of cations in electrolysis. Cations at lower positions of the electrochemical series are more readily discharged because they are stronger oxidizing agents (i.e. they gain electrons more readily).

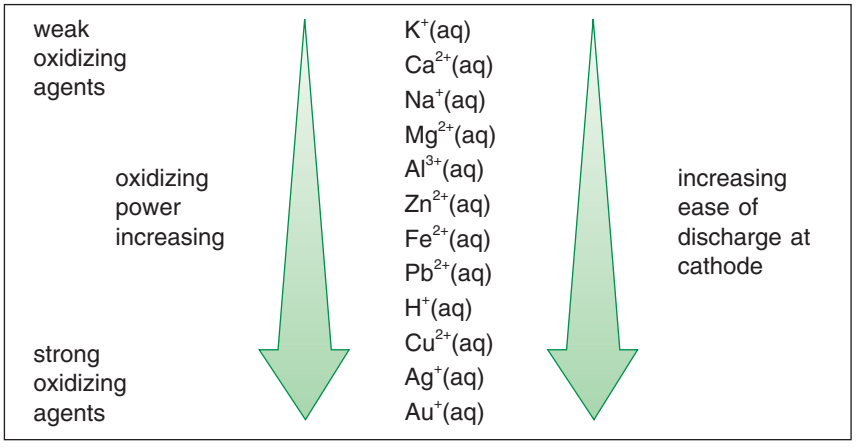
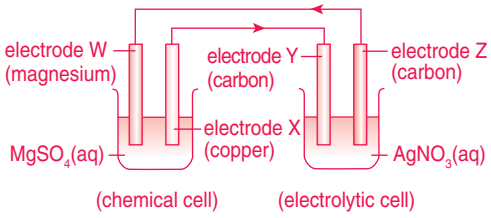


Fig. 22.5 Order of discharge of cations

N6 Connecting a chemical cell to an electrolytic cell

Examination questions may show a set-up connecting a chemical cell to an electrolytic cell. Students need to distinguish clearly between the changes that occur in the chemical cell and the electrolytic cell.

Example 1



Electrode W loses electrons and forms magnesium ions. The electrons flow through the electric wires to electrode Z.

Electrode Z: $Ag^+(aq) + e^- \longrightarrow Ag(s)$

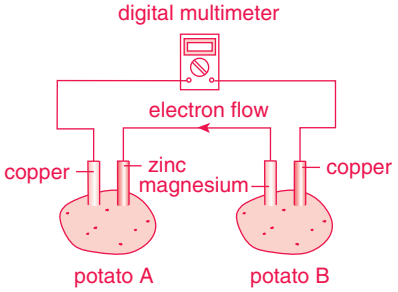
Electrode Y: $4OH^-(aq) \longrightarrow O_2(g) + 2H_2O + 4e^-$

Oxidation occurs at electrode W and it is the anode.

Reduction occurs at electrode Z and it is the cathode.

Example 2

Two competing potato cells



In the above set-up, two competing potato cells are connected together. The more powerful cell (i.e. cell B) drives the weaker cell (i.e. cell A) into electrolysis.

Consider the electrolysis of very dilute sodium chloride solution. A hydrogen ion is below a sodium ion in the electrochemical series. That means a hydrogen ion is a stronger oxidizing agent than a sodium ion. Hydrogen ions gain electrons more readily. Thus, they are preferentially discharged at the cathode.

If a cation is above the hydrogen ion in the electrochemical series, a hydrogen ion will discharge preferentially to form hydrogen gas.

If a cation is below the hydrogen ion in the electrochemical series, the cation will discharge preferentially to form a metal.

Order of discharge of anions

Fig. 22.6 shows the order of discharge of anions. Anions at higher positions of the electrochemical series are more readily discharged because they are stronger reducing agents (i.e. they lose electrons more readily).

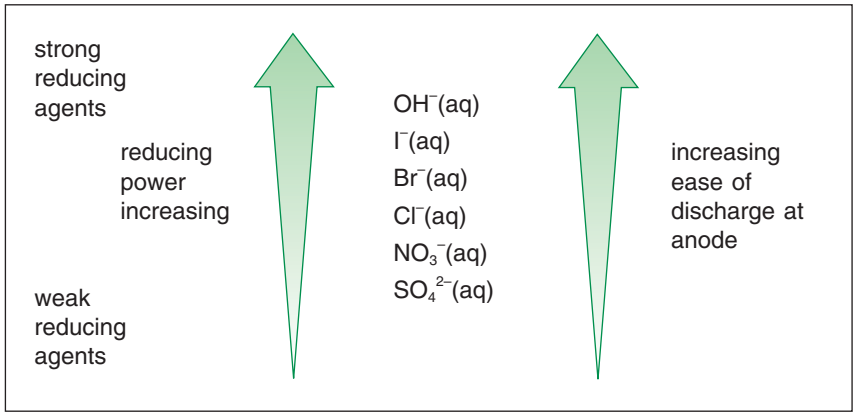


Fig. 22.6 Order of discharge of anions

Consider the electrolysis of very dilute sodium chloride solution. A hydroxide ion is above a chloride ion in the electrochemical series. That means a hydroxide ion is a stronger reducing agent than a chloride ion. Hydroxide ions lose electrons more readily. Thus, they are preferentially discharged at the anode. N6

22.3

Investigating factors affecting the order of discharge of ions during electrolysis — effect of concentration of ions in the solution.

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

The concentration of the dilute sodium chloride solution used is about 2 mol dm^{-3} while that of the concentrated sodium chloride solution is higher than 2 mol dm^{-3}.

Animation

Electrolysis of concentrated sodium chloride solution using carbon electrodes

22.8 The effect of concentration of ions in the solution and the order of discharge of ions

Electrolysis of dilute or concentrated sodium chloride solution using carbon electrodes

Fig. 22.7 shows the electrolysis of dilute or concentrated sodium chloride solution.

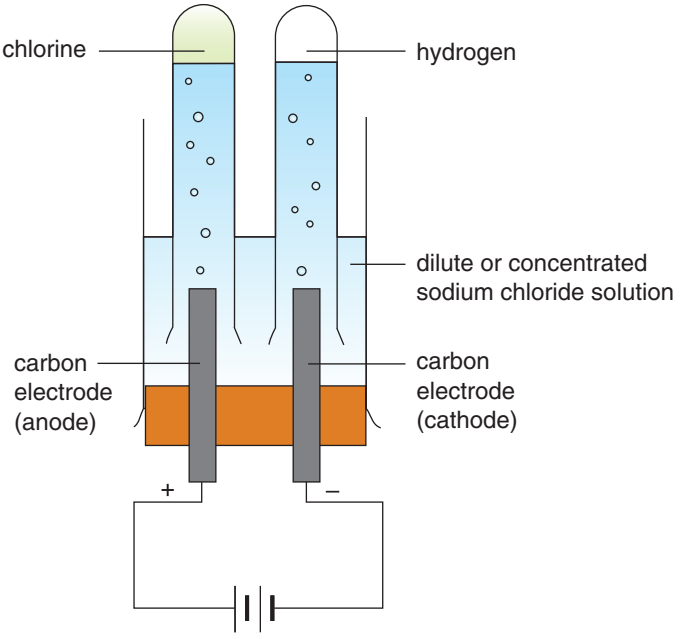


Fig. 22.7 Electrolysis of dilute or concentrated sodium chloride solution N7

N7 Students should be able to draw a diagram of the set-up for the electrolysis of sodium chloride solution.

In dilute or concentrated sodium chloride solution, there are four kinds of ions: $Na^+(aq)$, $Cl^-(aq)$, $H^+(aq)$ and $OH^-(aq)$.

Ions attracted to the anode	$Cl^-(aq)$, $OH^-(aq)$
Ions attracted to the cathode	$Na^+(aq)$, $H^+(aq)$

At the anode

Platinum electrodes should not be used here because chloride gas attacks platinum.

Chloride ions are preferentially discharged (oxidized) to form chlorine gas.

CE 2010 Paper 1 Q11(f)(i)

CE 2005 Paper 1 Q9(b)(i)

$$2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-$$

We have learnt that a hydroxide ion is a stronger reducing agent than a chloride ion. Thus, hydroxide ions should be discharged preferentially. However, at higher concentrations of chloride ions, the discharge of chloride ions becomes more favourable.

At the cathode

N8 When explaining which ion would be preferentially discharged at the anode / cathode, first state all kinds of ions that are attracted to the anode / cathode. This makes the answer more complete.

Notice that hydrogen ions / chloride ions are preferentially discharged, NOT hydrogen / chlorine.

A hydrogen ion is a stronger oxidizing agent than a sodium ion. Thus, hydrogen ions are preferentially discharged (reduced) to form hydrogen gas. **N8**

CE 2010 Paper 1 Q11(f)(ii)

CE 2005 Paper 1 Q9(a)(i)

$$2H^+(aq) + 2e^- \longrightarrow H_2(g)$$

Theoretically, the volume of hydrogen: the volume of chlorine is 1:1. In reality, the volume of chlorine collected is less than that of hydrogen as chlorine is quite soluble in water.

CE 2011 Paper 2 Q37

Overall cell reaction

AL 2009 Paper 1 Q3(a)(iv)

$$2H^+(aq) + 2Cl^-(aq) \longrightarrow H_2(g) + Cl_2(g)$$

When chlorine dissolves in water, it reacts reversibly with water, forming hydrochloric acid and hypochlorous acid (HOCl). The solution formed is acidic.

$$Cl_2(g) + H_2O(l) \rightleftharpoons HCl(aq) + HOCl(aq)$$

Changes in the solution

The chlorine gas formed at the anode dissolves in the solution. The solution there becomes acidic.

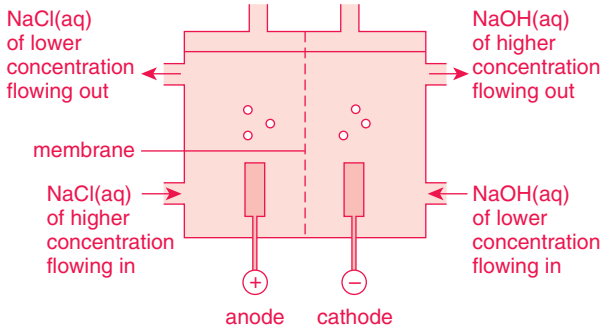
Water dissociates continuously to replace the hydrogen ions discharged at the cathode. Thus, there is a build-up of hydroxide ions near to the cathode. The solution there becomes alkaline.

Hydrogen ions and chloride ions are consumed in the electrolysis. Sodium ions and hydroxide ions remain in the solution. Eventually, the solution becomes sodium hydroxide solution. **N9**

CE 2010 Paper 1 Q11(f)(iii)
AL 2009 Paper 1 Q3(a)(iv)

N9 Electrolytic cell used in the manufacture of sodium hydroxide

The following diagram shows the electrolytic cell used in the manufacture of sodium hydroxide:



Choride ions are discharged at the anode to form chlorine gas. Hydrogen ions are discharged at the cathode to form hydrogen gas. Hydroxide ions remain in the cathodic compartment.

Unreacted sodium ions in the anodic compartment pass through the membrane to the cathodic compartment. Thus, sodium hydroxide solution of higher concentration is obtained.

Example 22.1

Q A very dilute sodium chloride solution is electrolyzed using inert electrodes for a long period of time.

a) State the expected observation at the cathode. Explain your answer with an appropriate ionic half-equation.

b) State ALL expected observations at the anode. Explain your answer with appropriate ionic half-equations.

c) Explain whether the resulting solution is acidic, alkaline or neutral.

CE 2011 Paper 1 Q10(a)

A a) Colourless gas bubbles (hydrogen) evolve.

A hydrogen ion is a stronger oxidizing agent than a sodium ion. Hydrogen ions are preferentially discharged (reduced) to form hydrogen gas.

$$2H^+(aq) + 2e^- \longrightarrow H_2(g)$$

b) Colourless gas bubbles (oxygen) evolve at the beginning.

A hydroxide ion is a stronger reducing agent than a chloride ion. Hydroxide ions are preferentially discharged (oxidized) to form oxygen gas.

$$4OH^-(aq) \longrightarrow O_2(g) + 2H_2O(l) + 4e^-$$

The sodium chloride solution becomes more concentrated as water is decomposed in the electrolysis. When the concentration of chloride ions is much higher than that of hydroxide ions, chloride ions are preferentially discharged (oxidized) to form chlorine gas.

$$2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-$$

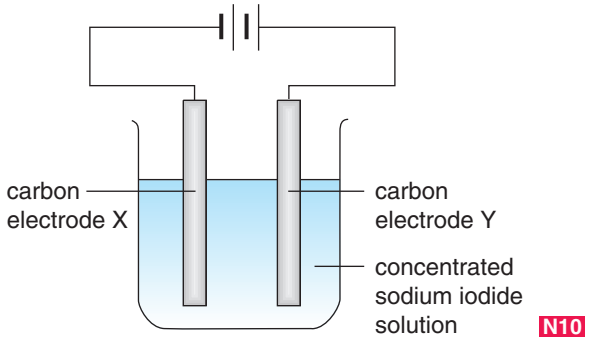
A greenish yellow gas evolves.

c) The resulting solution is alkaline. After electrolysis the concentration of $OH^-(aq)$ ions is higher than that of $H^+(aq)$ ions.

Practice 22.2

P22.2 Refer to Teacher's Resources Book page 56.

Electricity is passed through concentrated sodium iodide solution in the set-up shown below for some time.



N10

a) List the ions that are attracted to electrodes X and Y respectively.

b) Explain, with the aid of ionic half-equations, the observations at

i) electrode X; and

ii) electrode Y.

DSE 2013 Paper 1B Q9(a),(b)(i)

N10 If a few drops of universal indicator solution is added, the solution near electrode X would gradually turn blue. Hydrogen ions are discharged at electrode X. The amount of $OH^-(aq)$ ions near to electrode X increases .

Strictly speaking, carbon and platinum are not completely inert. When oxygen is given off at a carbon electrode, the product is *contaminated* with carbon dioxide due to the oxidation of carbon. Besides, chlorine gas attacks platinum.

22.9 The nature of electrodes and the order of discharge of ions

The electrodes we have used so far are made of inert substances such as carbon or platinum. They do not react with the electrolyte or the products of electrolysis. However, the nature of electrodes can affect the order of discharge of ions.

22.4

Investigating factors affecting the order of discharge of ions during electrolysis — effect of the nature of electrodes.

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

Electrolysis of dilute copper(II) sulphate solution using carbon electrodes

Fig. 22.8 shows the electrolysis of dilute copper(II) sulphate solution using carbon electrodes.

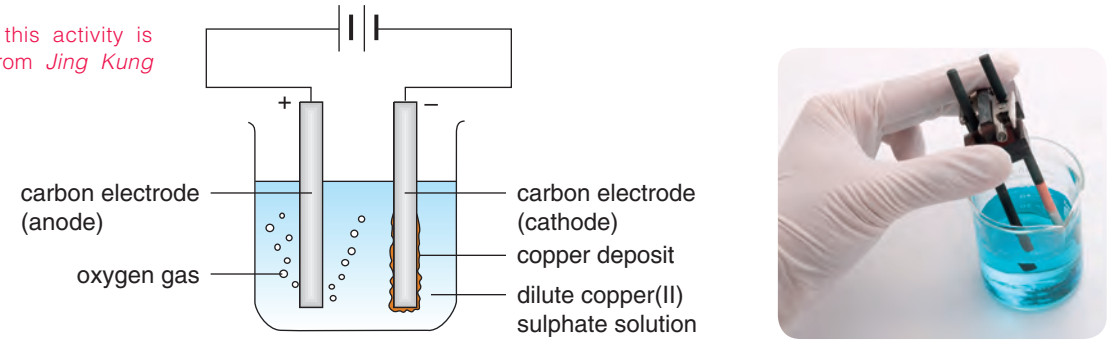


Fig. 22.8 Electrolysis of dilute copper(II) sulphate solution using carbon electrodes

In dilute copper(II) sulphate solution, there are four kinds of ions: $Cu^{2+}(aq)$, $SO_4^{2-}(aq)$, $H^+(aq)$ and $OH^-(aq)$.

Ions attracted to the anode	$SO_4^{2-}(aq)$, $OH^-(aq)$
Ions attracted to the cathode	$Cu^{2+}(aq)$, $H^+(aq)$

At the anode

Hydroxide ions are preferentially discharged (oxidized) to form oxygen gas.

$$4OH^-(aq) \longrightarrow O_2(g) + 2H_2O(l) + 4e^-$$

DSE 2014 Paper 1B Q8(a)(ii),(b)(i)

contaminate 污染

At the cathode

A copper(II) ion is a stronger oxidizing agent than a hydrogen ion. Hence copper(II) ions are preferentially discharged (reduced) to form a deposit of copper on the cathode.

$$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$$

Overall cell reaction

$$2Cu^{2+}(aq) + 4OH^-(aq) \longrightarrow 2Cu(s) + O_2(g) + 2H_2O(l)$$

CE 2008 Paper 2 Q48

Changes in the solution

Water dissociates continuously to replace the hydroxide ions discharged at the anode. Thus, there is a build-up of hydrogen ions near to the anode. The solution there becomes acidic. The blue colour of the solution fades gradually because the concentration of copper(II) ions decreases.

Copper(II) ions and hydroxide ions are consumed in the electrolysis. Hydrogen ions and sulphate ions remain in the solution. Thus, the solution eventually becomes sulphuric acid.

Electrolysis of dilute copper(II) sulphate solution using copper electrodes N11

Fig. 22.9 shows the electrolysis of dilute copper(II) sulphate solution using copper electrodes.

CE 2010 Paper 2 Q34

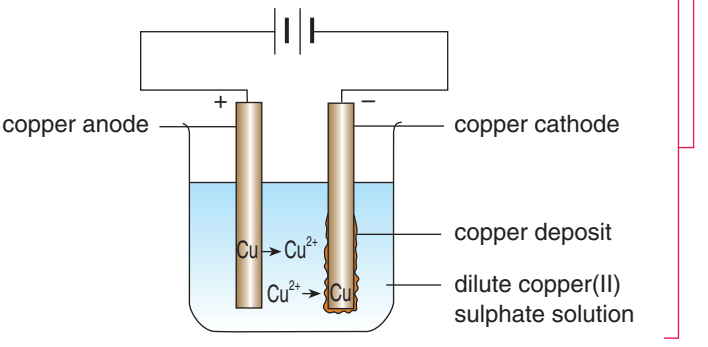


Fig. 22.9 Electrolysis of dilute copper(II) sulphate solution using copper electrodes

N11 When copper or metals more reactive than copper are made the anode in electrolysis, they usually dissolve to form ions. These electrodes are called active electrodes as they take part in the electrolysis.

For example, during the electrolysis of dilute zinc nitrate solution with zinc as the anode, the zinc anode would dissolve to form zinc ions:

$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

The electrolyte contains the following four kinds of ions: $Cu^{2+}(aq)$, $SO_4^{2-}(aq)$, $H^+(aq)$ and $OH^-(aq)$.

Ions attracted to the anode	$SO_4^{2-}(aq)$, $OH^-(aq)$
Ions attracted to the cathode	$Cu^{2+}(aq)$, $H^+(aq)$

At the anode

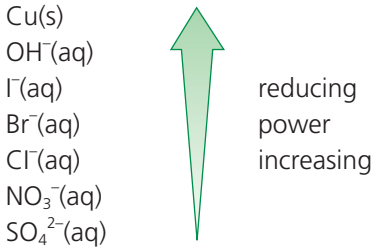
Compare this reaction with the reaction occurred at the anode in the electrolysis of dilute copper(II) sulphate solution using carbon electrodes.

Copper is a stronger reducing agent than hydroxide ion and sulphate ion. The copper anode dissolves to form copper(II) ions (oxidized).

$$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$$

At the cathode

According to the electrochemical series shown below, the position of copper is higher than that of the hydroxide ion and the sulphate ion. Thus, copper is a stronger reducing agent.



Copper(II) ions are preferentially discharged (reduced) to form a deposit of copper on the cathode.

$$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$$

Overall cell reaction

$$\underset{\text{(anode)}}{Cu(s)} \longrightarrow \underset{\text{(cathode)}}{Cu(s)}$$

Changes in the solution

The net effect is the transfer of copper from the anode to the cathode. The rate at which copper deposits on the cathode is equal to the rate at which the copper anode dissolves:

increase in mass of cathode = decrease in mass of anode

The concentration of copper(II) ions in the electrolyte remains the same. The blue colour of the solution does not change.

CE 2010 Paper 2 Q34

N12 Refer to Teacher's Resources Book page 31.

Table 22.2 summarizes the products of electrolyzing dilute copper(II) sulphate solution using different electrodes. N12

Table 22.2

Products of electrolyzing dilute copper(II) sulphate solution using different electrodes

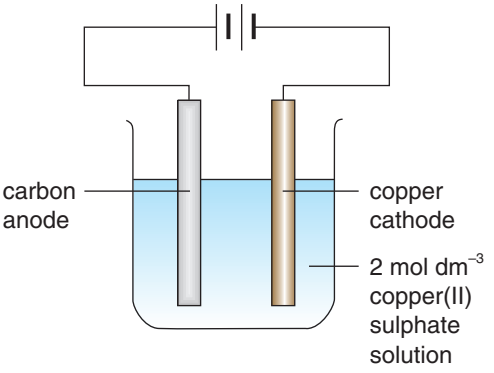
Material of		Product at		Change in the solution
anode	cathode	anode	cathode	
carbon	carbon	oxygen gas	copper	becomes sulphuric acid
copper	copper	copper(II) ions	copper	remains the same

CE 2005 Paper 2 Q31

Practice 22.3

P22.3 Refer to Teacher's Resources Book page 56.

A 2 mol dm^{-3} copper(II) sulphate solution is electrolyzed using a carbon anode and a copper cathode.



a) List the ions that are attracted to the carbon anode and copper cathode respectively.

b) i) Describe and explain what will happen at each electrode.

ii) Write an ionic half-equation for the reaction that occurs at each electrode.

c) What will happen to the pH of the electrolyte after some time? Explain your answer.

d) What will be the colour change of the electrolyte after some time? Explain your answer.

Electrolysis of concentrated sodium chloride solution using a mercury cathode

Fig. 22.10 shows the electrolysis of concentrated sodium chloride solution using a carbon anode and a mercury cathode.

We will discuss the industrial use of this electrolysis process in Topic 13 Industrial Chemistry.

If the wire is not covered, electrolysis of water will occur.

Animation
Manufacture of chlorine by electrolysis of brine — Mercury electrolytic cell

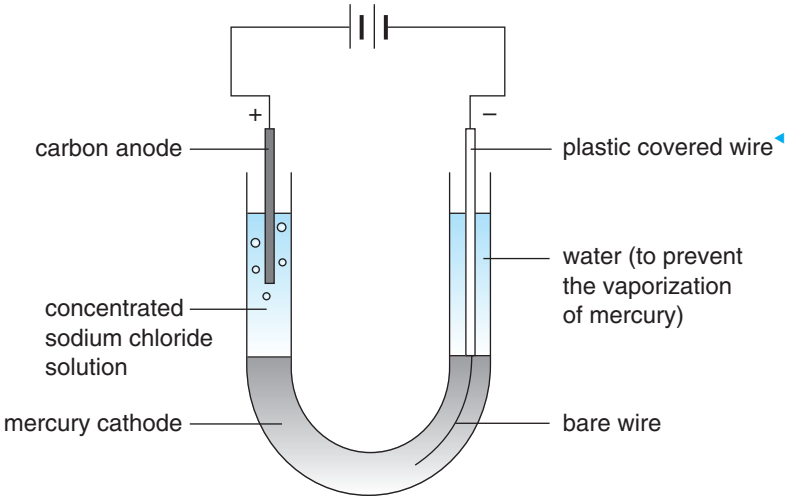


Fig. 22.10 Electrolysis of concentrated sodium chloride solution using a carbon anode and a mercury cathode

In concentrated sodium chloride solution, there are four kinds of ions: $Na^+(aq)$, $Cl^-(aq)$, $H^+(aq)$ and $OH^-(aq)$.

Ions attacted to the anode	$Cl^-(aq)$, $OH^-(aq)$
Ions attacted to the cathode	$Na^+(aq)$, $H^+(aq)$

At the anode

The concentration of chloride ions is much higher than that of hydroxide ions. Chloride ions are preferentially discharged (oxidized) to form chlorine gas.

$$2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-$$

At the cathode

A sodium ion is a weaker oxidizing agent than a hydrogen ion. However, if mercury is used as the cathode, sodium ions are preferentially discharged (reduced) to form sodium metal. The metal formed dissolves in the mercury cathode to form an alloy called **sodium amalgam**.

CE 2010 Paper 2 Q32

$$Na^+(aq) + e^- + Hg(l) \longrightarrow \underset{\text{sodium amalgam}}{Na/Hg(l)}$$

The sodium amalgam then moves towards the water. The sodium reacts with water to form sodium hydroxide and hydrogen.

This reaction is usually not carried out in a school laboratory.

$$2Na/Hg(l) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g) + 2Hg(l)$$

Overall cell reaction

$$2Na^+(aq) + 2Cl^-(aq) + 2Hg(l) \longrightarrow 2Na/Hg(l) + Cl_2(g)$$

Changes in the solution

Sodium ions and chloride ions are consumed in the electrolysis. Thus, the sodium chloride solution becomes more and more dilute.

Table 22.3 summarizes the products of electrolyzing sodium chloride solutions of different concentrations using different electrodes.

Table 22.3

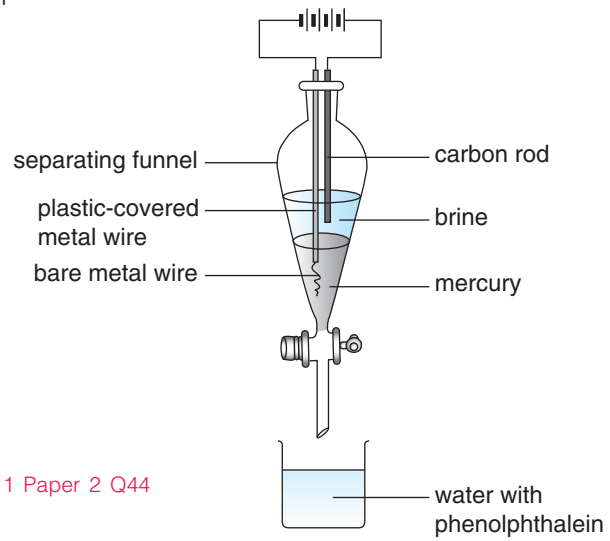
Products of electrolyzing sodium chloride solutions of different concentrations using different electrodes

Sodium chloride solution of different concentrations	Material of		Product at		Change in the solution
	anode	cathode	anode	cathode	
very dilute	carbon	carbon	oxygen gas	hydrogen gas	becomes more concentrated
dilute / concentrated	carbon	carbon	chlorine gas	hydrogen gas	becomes sodium hydroxide solution
concentrated	carbon	mercury	chlorine gas	sodium	becomes more dilute

sodium amalgam 鈉汞齊

Practice 22.4 P22.4 Refer to Teacher's Resources Book page 57.

The following diagram shows the set-up of an experiment.



CE 2011 Paper 2 Q44

After starting the experiment for some time, the tap of the separating funnel is opened to run some mercury into the beaker containing water with phenolphthalein. The water with phenolphthalein turns red.

a) Decide whether the carbon rod is the anode or the cathode.

CE 2011 Paper 2 Q44

b) Write ionic half-equations for the reactions that occur at the carbon rod and the mercury electrode respectively.

c) With the aid of an appropriate equation, explain why the water with phenolphthalein turns red.

CE 2011 Paper 2 Q44

22.10 Industrial uses of electrolysis

Refining of copper

When copper is first extracted from its ore, it is about 99% pure. The impurities, mostly silver, gold, platinum, iron and zinc, reduce the electrical conductivity of copper significantly. Hence copper must be further *refined* before it is used to manufacture electrical wires. Fig. 22.11 illustrates how this is done.

The copper used in electrical wires must be very pure.

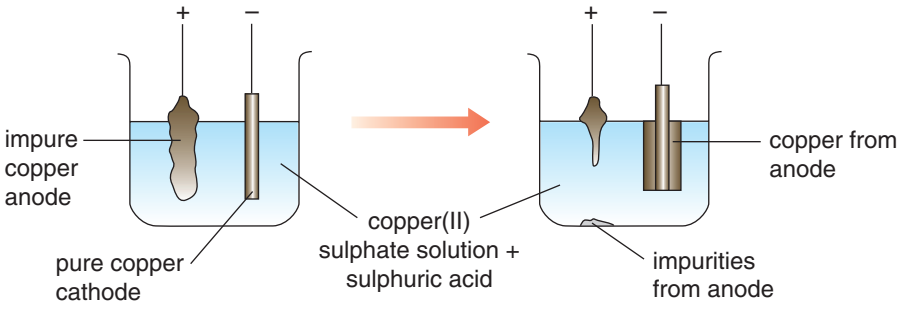


Fig. 22.11 Refining of copper

refining 精煉

Sulphuric acid increases the electrical conductivity of the electrolyte.



Fig. 22.12 Refining of copper in a factory



Fig. 22.13 These copper cathodes from the refining cells are ready for processing into copper wires and other copper products

The electrolytic cell contains a solution of copper(II) sulphate and sulphuric acid as the electrolyte. The impure copper becomes the anode. The cathode is a thin sheet of very pure copper (Fig. 22.13).

At the anode

CE 2007 Paper 1 Q11(b),(c),(d)

Iron and zinc form ions more readily than copper. When the cell is operating at the correct voltage, iron and zinc in the anode give up electrons first. Then copper gives up electrons to form copper(II) ions. Impurities such as silver, gold and platinum settle at the bottom of the container.

$$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$$

$$Fe(s) \longrightarrow Fe^{2+}(aq) + 2e^-$$

$$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$$

At the cathode

A copper(II) ion is a stronger oxidizing agent than zinc ion and iron(II) ion. Thus, copper(II) ions are discharged to form copper.

$$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$$

Overall cell reaction

$$\underset{\text{(anode)}}{Cu(s)} \longrightarrow \underset{\text{(cathode)}}{Cu(s)}$$

During the refining process, the copper is gradually transferred from the anode to the cathode. The concentration of copper(II) ions in the electrolyte drops gradually. This is because at the anode, iron and zinc readily dissolve as ions, while at the cathode, copper(II) ions are always preferentially discharged.

This process produces copper that is 99.95% pure. High-purity gold can also be produced.

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

22.5
Electroplating with nickel.

Video
Electroplating with nickel

Animation
Electroplating

Electroplating

CE 2009 Paper 1 Q13
ASL 2005 Q7(II)(a)(ii)

Metals that readily corrode can be protected by **electroplating** or applying a thin plating of a metal that resists corrosion. Examples are chrome-plated steel car bumpers (Fig. 22.14), silver-plated jewellery and decorative items and 'tin' cans (which are actually steel cans with a thin coating of tin). N13

ASL 2006 Q13(b)(II)(i)



Fig. 22.14 Vintage cars have many chrome-plated parts

In this process, we clean the object to be plated and make it the cathode. The plating metal is made the anode. An aqueous solution of a salt of the plating metal is used as the electrolyte (Fig. 22.15). The plating metal is gradually transferred onto the object.

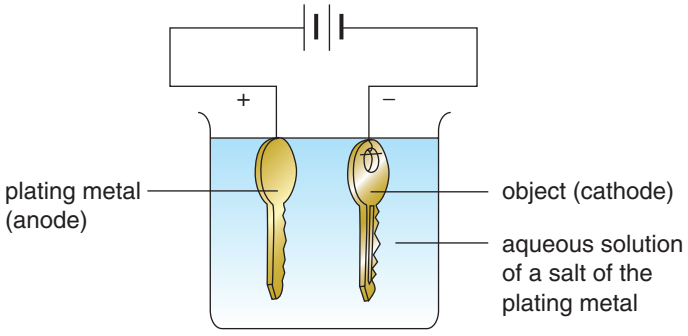
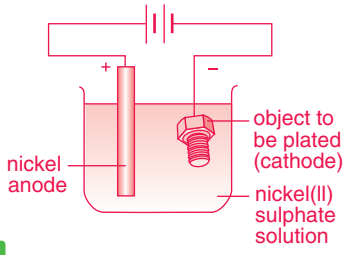


Fig. 22.15 Experimental set-up for electroplating

CE 2009 Paper 1 Q13
DSE 2014 Paper 1A Q11

N13 Examination questions often ask about the set-up for electroplating nickel on an object.

Anode	Cathode	Electrolyte
nickel	object to be plated	nickel(II) sulphate solution



Do you know

Clean surface for electroplating

In the industrial electroplating of metals, it is most important to obtain a clean surface first. A small amount of contaminants would prevent the layer of plating metal from sticking firmly to the metal to be plated. The metal to be electroplated is usually cleaned using solvents and hot alkaline solutions.

electroplating 電鍍

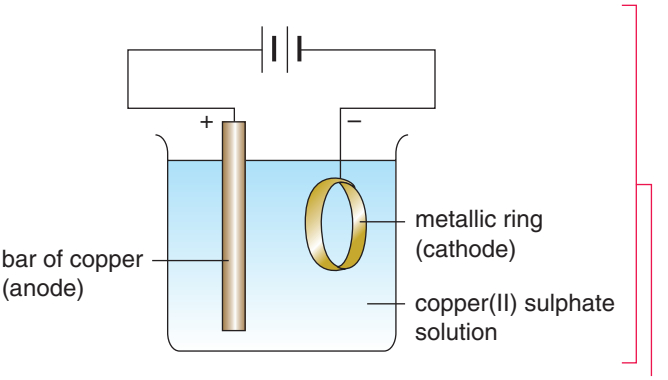
Example 22.2

Q A worker used the set-up shown on the right to coat a layer of copper on a metallic ring by electrolysis. The worker used a very high voltage. Some bubbles were found to form on the ring and the copper layer flaked off easily.

a) Explain why copper can be coated on the metallic ring by electrolysis.
DSE 2012 Paper 1B Q5(b)(i)

b) Write an ionic half-equation for the reaction taking place at the metallic ring.
CE 2007 Paper 2 Q45

c) Suggest what the bubbles were, and explain why the copper layer flaked off easily.
DSE 2012 Paper 1B Q5(b)(ii)



DSE 2012 Paper 1B Q5(c)

A a) A copper(II) ion is a stronger oxidizing agent than a hydrogen ion. Hence copper(II) ions are preferentially discharged (reduced) to form a deposit of copper on the metallic ring.

b) $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$

c) Hydrogen gas

The hydrogen gas bubbles hinder the deposition of copper on the surface of the metallic ring.

Practice 22.5

P22.5 Refer to Teacher's Resources Book page 57.

An electroplating factory produces nickel-plated plastic turning knobs on radios. In the manufacturing process, the plastic knobs are first coated with copper and then electroplated with nickel.

a) Why are the plastic knobs first coated with copper before electroplating? ASL 2005 Q7(II)(a)(i)

b) Write an ionic half-equation for the reaction that occurs at a turning knob. ASL 2005 Q7(II)(b)

c) Draw a labelled diagram to show the laboratory set-up for the above electroplating process.



Fig. 22.16 Harmful chemical waste discharged into a river

22.11 Environmental impact of the electroplating industry

The electroplating industry produces many poisonous waste by-products. If these poisonous waste products are discharged to the environment without treatment, they will cause serious pollution problems (Fig. 22.16).

Pollutants from the electroplating industry include:

- acids and alkalis;
- compounds of heavy metals; and
- *cyanides*.

Acids and alkalis cause changes in the pH of water and affect water life. Cyanides are highly toxic. They give *lethal* hydrogen cyanide gas when reacted with acids. Ions of heavy metals (such as nickel, cadmium, chromium and mercury) are readily absorbed into the bodies of living organisms such as shellfish and plants. Humans may get poisoned by eating them.

Methods to control pollution from the electroplating industry

Reducing the volume of waste solutions

Design better rinsing systems to reduce water usage.

Recovering useful materials from waste solutions

In most electroplating processes, a layer of nickel is first plated as a base on which the ultimate surface layer is deposited. As a result, nickel is commonly found in the waste solutions. It can be recovered by treating the waste solutions.

cyanides 氰化物　　lethal 致命的

Treating effluents before discharge

We have discussed this in Topic 4 Acids and Bases.

- Controlling the pH of *effluents*
- Treatment of heavy metal compounds

Most heavy metal hydroxides are insoluble in water. We can add sodium hydroxide solution to the effluents. The solution reacts with the heavy metal ions to form insoluble metal hydroxides. The solid can then be filtered off.

- Treatment of chromium waste

Chromium(III) compounds are less toxic than chromium(VI) compounds.

The chromium-plating solution contains chromium(VI) compounds. These are very toxic. They are normally reduced to chromium(III) compounds by sodium sulphite. Sodium hydroxide solution is then added to the chromium(III) compounds to form solid chromium(III) hydroxide. The solid is then filtered off.

CE 2005 Paper 1 Q4(c)
ASL 2006 Q13(b)(II)(v)

effluent 廢水

Have you mastered?

Key terms

electrolysis 電解 101 electrolytic cell 電解池 101 electrolyte 電解質 101 anode 陽極 101

cathode 陰極 101 discharge preferentially 優先放電 104 mercury cathode 汞陰極 117

sodium amalgam 鈉汞齊 118 refining of copper 銅的精煉 119 electroplating 電鍍 121

effluent 廢水 124

Checklist

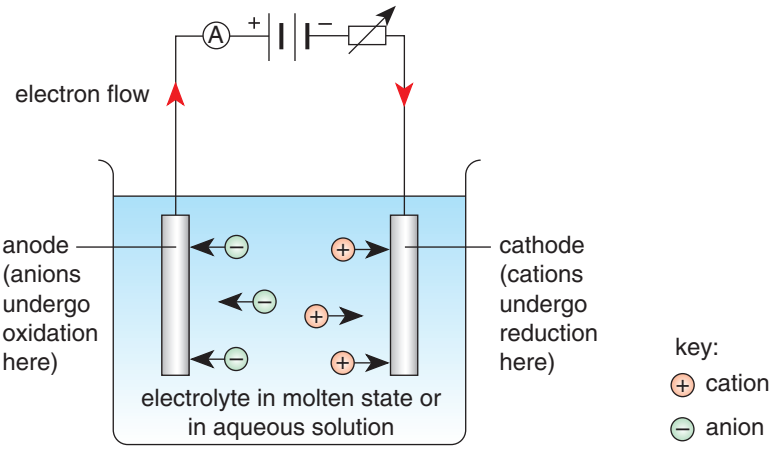
After studying this unit, you should be able to

- ☐ predict the products at each electrode of an electrolytic cell with reference to the factors affecting the preferential discharge of ions;
- ☐ describe the anodic and cathodic reactions, overall reaction and observable changes in an electrolytic cell;
- ☐ understand the principles of refining of copper and electroplating;
- ☐ describe the anodic and cathodic reactions, overall reactions and observable changes in the refining of copper and electroplating;
- ☐ understand the environmental impact of the electroplating industry.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Electrolysis is the chemical reaction that occurs when electricity passes through an electrolyte in molten state or in aqueous solution.



2 An electrolytic cell is a cell in which electrolysis occurs. The following table summarizes some facts concerning the electrodes of an electrolytic cell.

Electrode	Connected to which electrode of d.c. supply	Ions discharged here	Process that occurs here
Anode	positive	anions	oxidation
Cathode	negative	cations	reduction

3 Factors affecting the order of discharge of ions during the electrolysis of aqueous solutions include:

a) the position of ions in the electrochemical series;

b) the concentration of ions in the solution; and

c) the nature of the electrodes used.

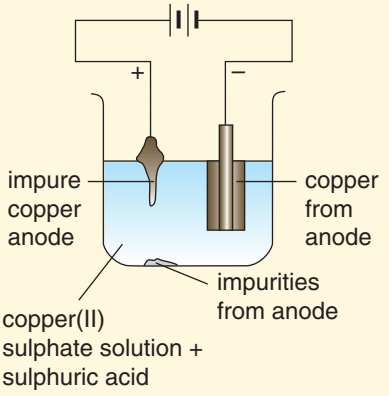
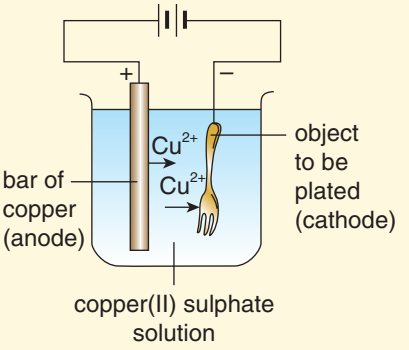
Have you mastered?

4 The following table summarizes the products of electrolysis of some molten liquids and aqueous solutions.

Substance		Material of		Product at		Change in the solution
		anode	cathode	anode	cathode	
Molten liquid	sodium chloride	carbon	carbon	chlorine gas	sodium	–
Aqueous solution	dilute sulphuric acid	platinum	platinum	oxygen gas	hydrogen gas	becomes more concentrated
	dilute hydrochloric acid	carbon	carbon	chlorine gas	hydrogen gas	becomes more dilute
	very dilute sodium chloride	carbon	carbon	oxygen gas	hydrogen gas	becomes more concentrated
	dilute sodium chloride	carbon	carbon	chlorine gas	hydrogen gas	becomes sodium hydroxide solution
	dilute sodium sulphate	carbon	carbon	oxygen gas	hydrogen gas	becomes more concentrated
	dilute sodium nitrate	carbon	carbon	oxygen gas	hydrogen gas	becomes more concentrated
	concentrated sodium chloride	carbon	carbon	chlorine gas	hydrogen gas	becomes sodium hydroxide solution
	concentrated sodium chloride	carbon	mercury	chlorine gas	sodium	becomes more dilute
	dilute copper(II) sulphate	carbon	carbon	oxygen gas	copper	becomes sulphuric acid
	dilute copper(II) sulphate	carbon	copper	oxygen gas	copper	becomes sulphuric acid
	dilute copper(II) sulphate	copper	copper	copper(II) ions	copper	remains the same
	dilute copper(II) sulphate	copper	carbon	copper(II) ions	copper	remains the same

Have you mastered?

5 The following table summarizes two industrial uses of electrolysis.

Use	refining of copper	electroplating
Anode	impure copper	plating metal
Cathode	pure copper	object to be plated
Electrolyte	copper(II) sulphate solution + sulphuric acid	aqueous solution of a salt of the plating metal
Cell reaction	**at the anode** $Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$ $Fe(s) \longrightarrow Fe^{2+}(aq) + 2e^-$ $Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$ **at the cathode** $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$ **overall cell reaction** $Cu(s) \longrightarrow Cu(s)$ (anode) (cathode)	example — plating an object with copper **at the anode** $Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$ **at the cathode** $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$ **overall cell reaction** $Cu(s) \longrightarrow Cu(s)$ (anode) (cathode)
Set-up		

6 Methods to control pollution from the electroplating industry include:

a) reducing the volume of waste solutions;

b) recovering useful materials from waste solutions; and

c) treating effluents before discharge.

Have you mastered?

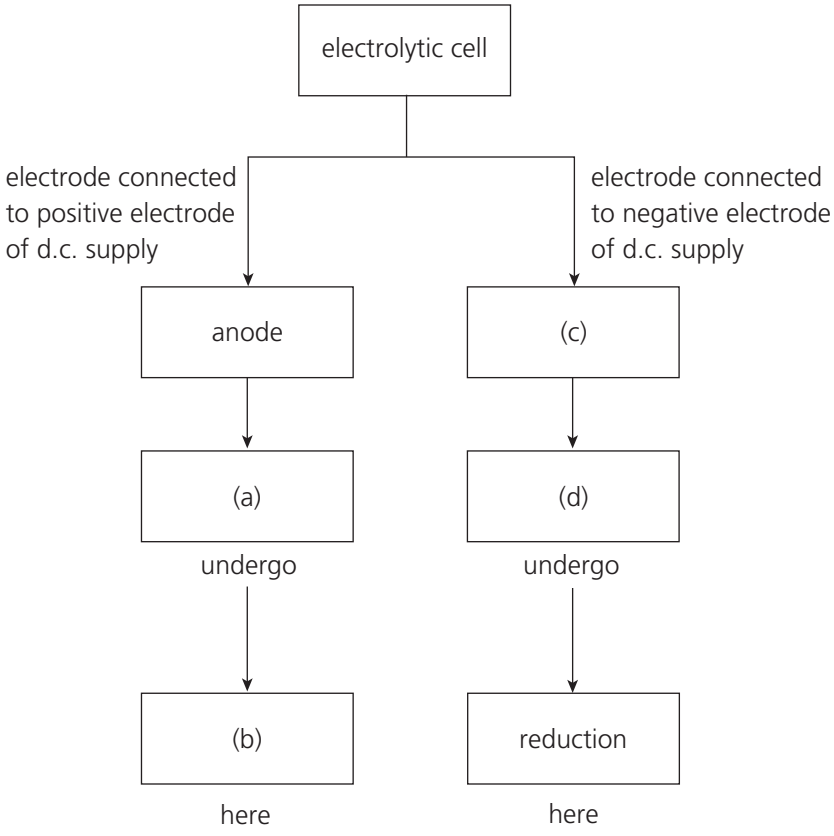
Unit Exercise

Refer to Teacher's Resources Book pages 58–63.

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



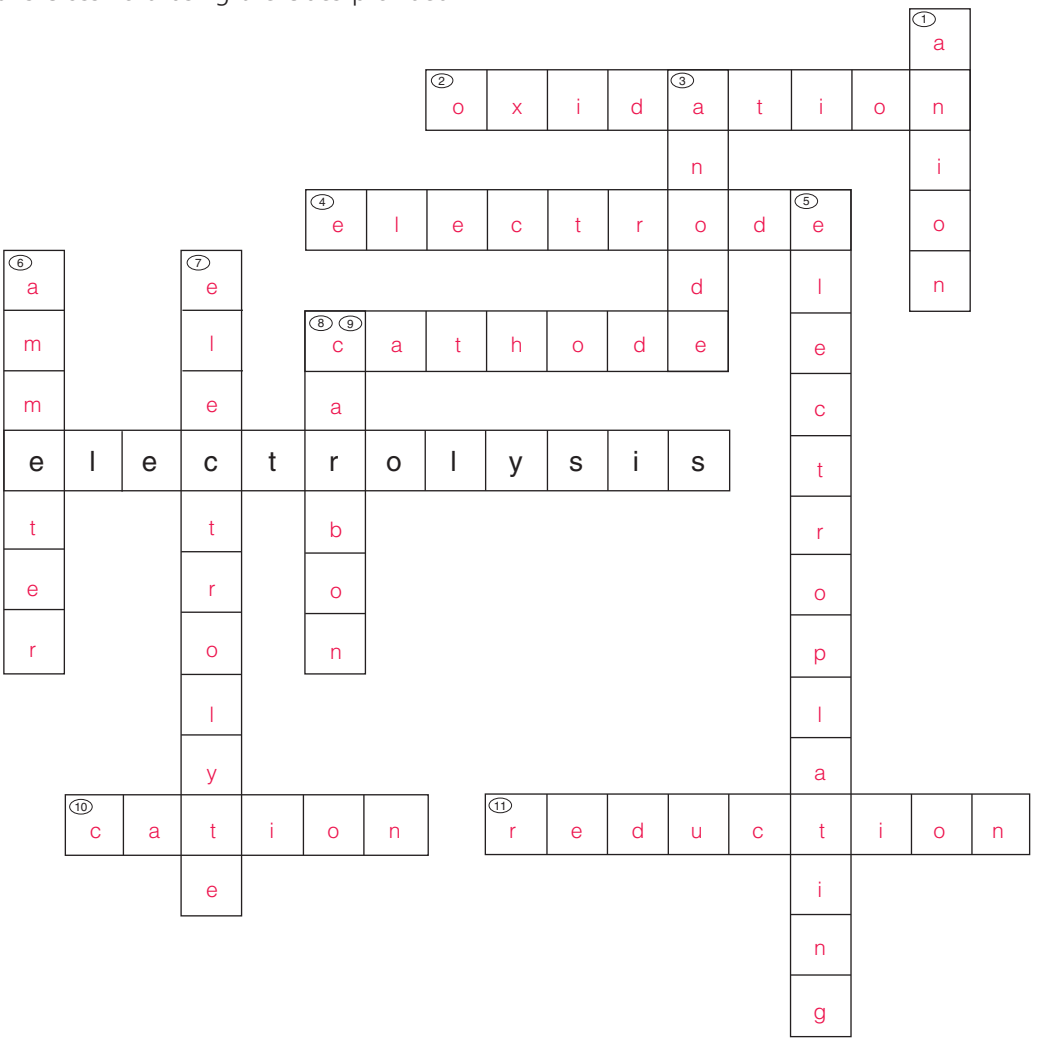
a) anions
b) oxidation
c) cathode
d) cations

2 Complete the following table on products of electrolysis of some aqueous solutions.

Solution	Material of		Product at		Change in the solution
	anode	cathode	anode	cathode	
Dilute sulphuric acid	platinum	platinum	oxygen gas	hydrogen gas	becomes more concentrated
Very dilute sodium chloride	carbon	carbon	oxygen gas	hydrogen gas	becomes more concentrated
Concentrated sodium chloride	carbon	carbon	chlorine gas	hydrogen gas	becomes sodium hydroxide solution
Concentrated sodium chloride	carbon	mercury	chlorine gas	sodium	becomes more dilute
Dilute copper(II) sulphate	carbon	carbon	oxygen gas	copper	becomes sulphuric acid
Dilute copper(II) sulphate	copper	carbon	copper(II) ions	copper	remains the same
Dilute copper(II) sulphate	copper	copper	copper(II) ions	copper	remains the same

3 Draw a labelled diagram of an electrolytic cell that can be used to plate an iron object with nickel.
Refer to Teacher's Resources Book page 58.

4 Complete this crossword using the clues provided.



Across

(2) A process in which a species loses electron(s).
(4) A pole where the electric current enters or leaves the electrolyte.
(8) The electrode of an electrolytic cell connected to the negative electrode of the d.c. supply.
(10) A positive ion.
(11) A process in which a species gains electron(s).

Down

(1) A negative ion.
(3) The electrode of an electrolytic cell connected to the positive electrode of the d.c. supply.
(5) The process of coating an object with a thin layer of metal by electrolysis.
(6) An instrument used to measure the electric current passing through a circuit.
(7) A substance which conducts electricity in molten state or aqueous solution, and is decomposed by electricity.
(9) A material used to make an inert electrode.

Part II Multiple choice questions

Refer to Teacher's Resources Book for explanations.

5 Which of the following statements concerning a working electrolytic cell is correct?

A Water must be present in the electrolytic cell.
B The electrolytic cell liberates electrical energy.
C The electrodes in the electrolytic cell must be metal.
(D) Redox reaction must be involved in the electrolytic cell.

(HKCEE, Paper 2, 2011, 35)

6 Which of the following would occur when a copper(II) sulphate solution is electrolyzed using carbon electrodes?

A The cathode dissolves.
B Hydrogen gas is produced.
(C) The pH of the solution decreases.
D The concentration of copper(II) ions in the solution increases.

7 The electrolysis of sodium sulphate solution using carbon electrodes produces changes in the solution around the electrodes. How will the pH change around the anode and the cathode?

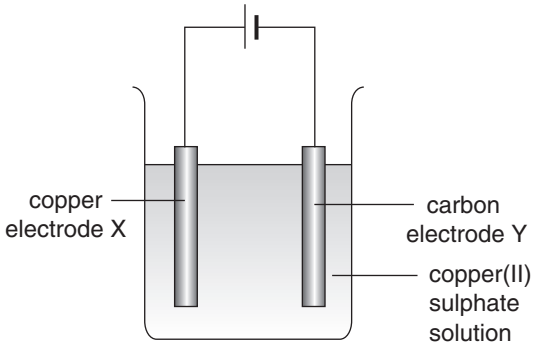
	pH around the anode	pH around the cathode
A	increases	increases
B	decreases	decreases
C	increases	decreases
(D)	decreases	increases

8 In the electrolysis of concentrated potassium chloride solution using carbon electrodes, gaseous products are liberated at both electrodes. After a short period of time, what is the theoretical volume ratio of the gas collected at the cathode to the gas collected at the anode?

(A) 1:1
B 1:2
C 1:4
D 2:1

(HKCEE, Paper 2, 2011, 37)

9 In an experiment to study the electrolysis of copper(II) sulphate solution, the set-up used is shown below:



Which of the following statements concerning the above experiment is correct?

A Reduction occurs at X.
B Hydrogen gas is evolved at Y.
C The pH of the solution increases gradually.
D The colour of the solution remains unchanged.

(HKCEE, Paper 2, 2010, 34)

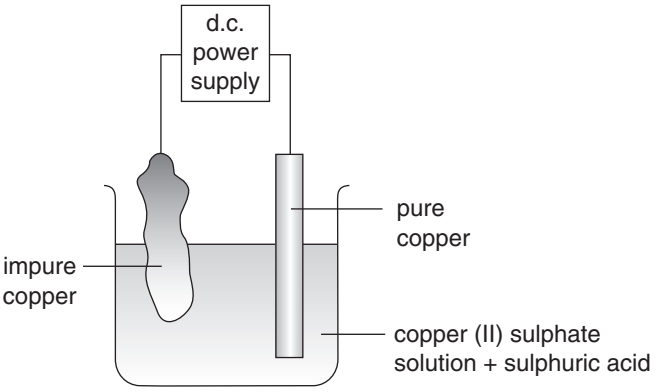
10 Potassium peroxodisulphate ($K_2S_2O_8$) can be obtained from the electrolysis of a saturated solution of potassium hydrogensulphate ($KHSO_4$).

Which of the following correctly describes the oxidation number of sulphur in $KHSO_4$, and the electrode at which $K_2S_2O_8$ is produced during the electrolysis?

	Oxidation number of S	Electrode at which $K_2S_2O_8$ is produced
A	+6	anode
B	+6	cathode
C	+4	anode
D	+4	cathode

(HKDSE, Paper 1A, 2013, 17)

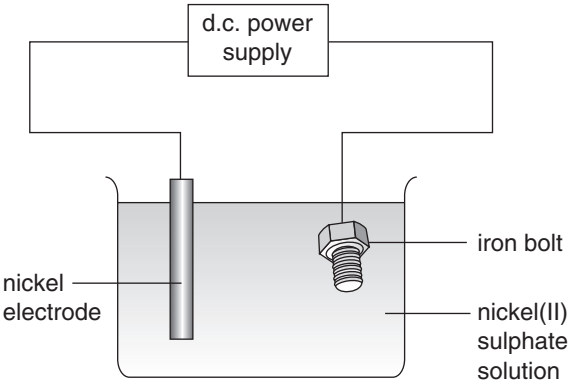
11 The following set-up is used to refine impure copper containing iron and zinc.



Which of the following statements about the set-up is correct?

A The impure copper acts as the cathode.
B Oxidation occurs at the anode.
C Iron and zinc in the impure copper undergo reduction at the cathode.
D The colour of the electrolyte remains unchanged.

12 An iron bolt is nickel-plated using the set-up shown below.



Which of the following statements concerning the experiment is INCORRECT?

A The mass of the iron bolt increases.
B The nickel electrode is the anode.
C The nickel(II) ions are reduced at the anode.
D The concentration of nickel(II) ions in the solution remains the same.

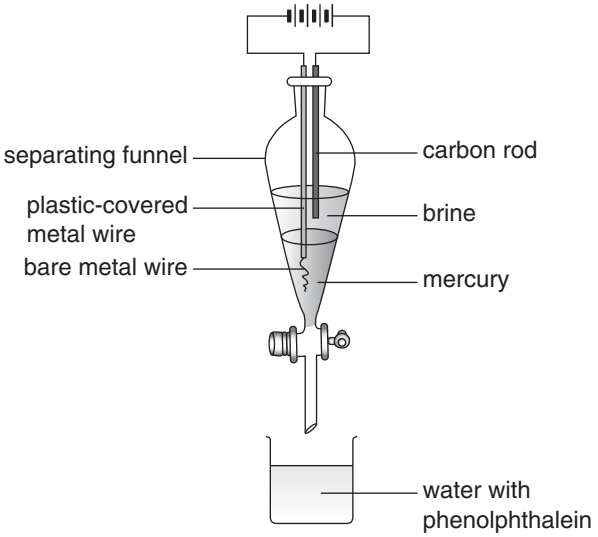
13 In the electrolysis of a copper(II) sulphate solution, carbon is used as the anode and copper as the cathode.

Which of the following statements about this electrolysis is / are correct?

(1) $O_2(g)$ is liberated at the anode.
(2) $H_2(g)$ is liberated at the cathode.
(3) The concentration of $Cu^{2+}(aq)$ ions in the solution remains unchanged.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

14 The following diagram shows the set-up of an experiment.



After some time, the tap of the separating funnel is opened to run some mercury from the separating funnel into a beaker conaining water with phenolphthalein. Which of the following statements concerning the experiment is / are correct?

(1) Mercury in the separating funnel can increase the electrical conductivity of the brine.
(2) The water with phenolphthalein turns red.
(3) The carbon rod acts as the cathode.

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2011, 44)

15 Upon electrolysis, which of the following solutions would give hydrogen at carbon cathode and oxygen at platinum anode?

(1) Very dilute sodium chloride solution
(2) Dilute copper(II) sulphate solution
(3) Concentrated potassium sulphate solution

A (1) only
B (2) only
(C) (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2008, 48)

16 Consider the electrolysis experiments using the following combinations of electrolyte solution, anode and cathode:

Electrolyte solution	Anode	Cathode
(1) copper(II) sulphate solution	copper	copper
(2) copper(II) chloride solution	graphite	graphite
(3) potassium sulphate solution	platinum	platinum

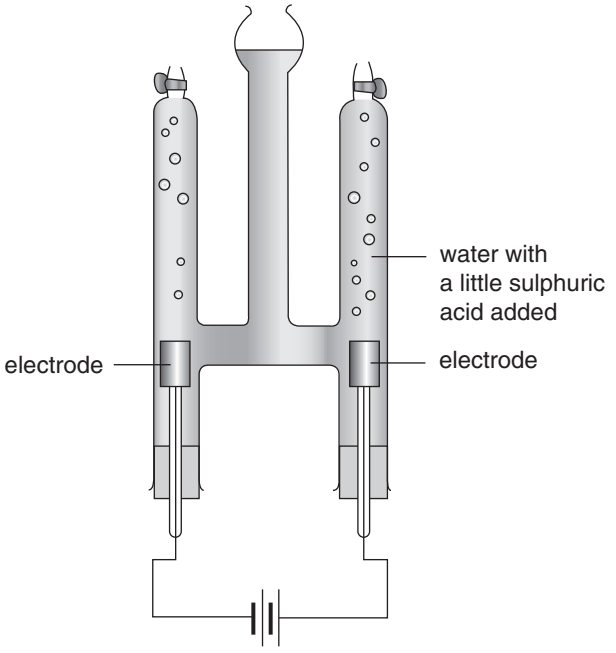
In which of these experiments will the concentration of the electrolyte solution remain UNCHANGED?

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKDSE, Practice paper 1A, 2012, 22)

Part III Structured questions

17 Water can be decomposed by electrolysis with the following set-up to give hydrogen and oxygen.



a) Explain why a little sulphuric acid has been added to the water used.

b) Suggest a suitable material for the electrodes.

c) Write ionic half-equations for the reactions that occur at the cathode and the anode respectively.

d) What is the theoretical volume ratio of the gas collected at the cathode to the gas collected at the anode?

e) What change would occur in the solution of the set-up after electrolysis? Explain your answer.

18 A very dilute sodium chloride solution is electrolyzed using inert electrodes for a long period of time.

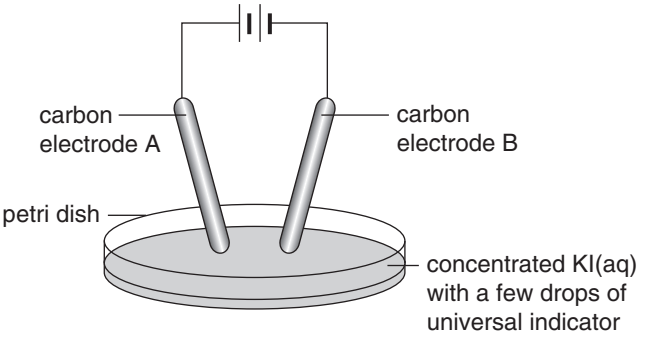
a) State the expected observation at the cathode. Explain your answer.

b) State ALL expected observations at the anode. Explain your answer.

c) Explain whether the resulting solution is acidic, alkaline or neutral.

(HKCEE, Paper 1, 2011, 10(a))

19 The diagram below shows the set-up used in an investigation on the electrolysis of concentrated potassium iodide solution:



a) State and explain the expected observation around carbon electrode A during the electrolysis.

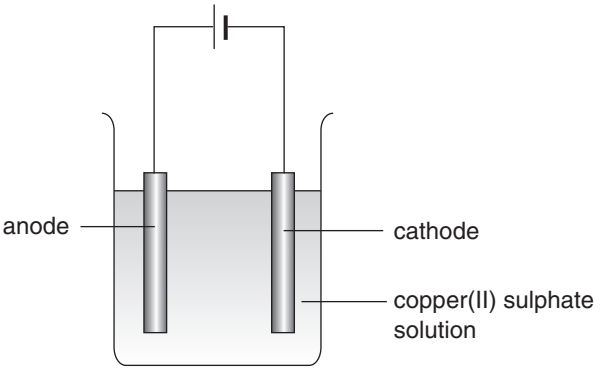
b) The solution near carbon electrode B gradually turned blue.

i) Explain this observation.

ii) Would there be any change in observation if carbon electrode B is replaced by a copper electrode in the investigation? Explain.

(HKDSE, Paper 1B, 2013, 9)

20 The electrolysis of 2 mol dm^{-3} copper(II) sulphate solution using pure copper electrodes was carried out using the following apparatus.



a) Name the particles responsible for the passage of current:

i) in the wires;

ii) in the copper(II) sulphate solution.

b) Write ionic half-equations for the reactions at the anode and cathode respectively.

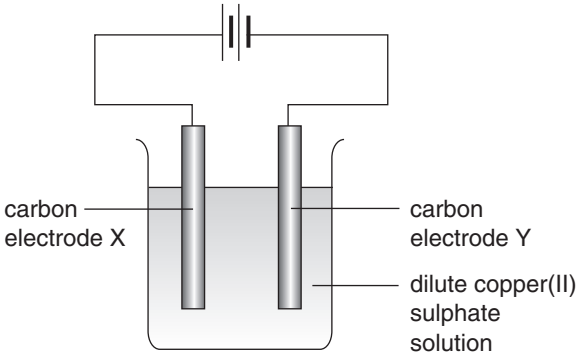
c) What was the concentration of the copper(II) sulphate solution at the end of the electrolysis? Underline the correct answer.

Less than 2 mol dm^{-3} Exactly 2 mol dm^{-3}

More than 2 mol dm^{-3}

(Edexcel GCE O Level, Paper 1, Jun. 2008, 5(a), (b), (d))

21 A dilute copper(II) sulphate solution was electrolyzed using carbon electrodes.

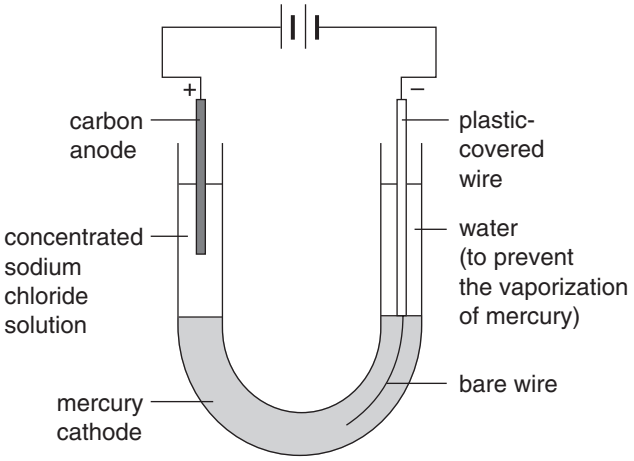


a) Describe and explain what would happen at each carbon electrode.

b) Write ionic half-equations for the reactions that occur at electrodes X and Y respectively.

c) What would be the change in the electrolyte after some time? Explain your answer.

22 The following diagram shows a set-up for the electrolysis of concentrated sodium chloride solution using a carbon anode and a mercury cathode.



a) Write ionic half-equations for the reactions that occur at

i) the carbon anode;

ii) the mercury cathode.

b) Two useful products can be obtained by treating the mercury in the above set-up.

i) Name these two products.

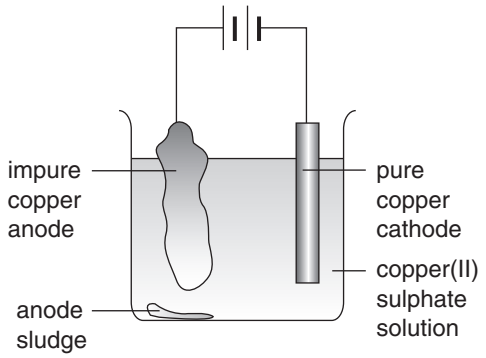
ii) Suggest how these products can be obtained. Give an appropriate equation.

c) What would happen to the concentrated sodium chloride solution after some time?

23 In a chemical plant, extraction of copper from its ores involves roasting copper(I) sulphide with air inside a high temperature furnace. Copper(I) sulphide reacts with oxygen in air according to the following equation:

$$Cu_2S + O_2 \longrightarrow 2Cu + SO_2$$

The copper so extracted contains impurities including metals such as silver, iron, zinc and gold. The impure copper is then purified by electrolysis as illustrated in the following diagram:



a) With reference to the reaction between copper(I) sulphide and oxygen, identify the species undergoing oxidation and the species undergoing reduction. Explain your answers in terms of changes in oxidation numbers.

b) Explain briefly how impure copper can be purified by electrolysis as illustrated in the diagram above.

c) Insoluble impurities deposit under the impure copper anode as 'anode sludge'. According to the information given, suggest what substances the anode sludge would contain. Explain your answer.

d) 'The concentration of copper(II) ions in copper(II) sulphate solution remains UNCHANGED in the above electrolysis.' Is this statement correct? Explain your answer.

(HKCEE, Paper 1, 2007, 11(a)–(d))

24 a) When silver nitrate solution, $AgNO_3$,is electrolyzed using inert electrodes, silver is deposited at the cathode and a colourless gas is evolved at the anode.

i) Write the ionic half-equation for the reaction at the cathode.

ii) Name the colourless gas evolved at the anode and give a test to confirm its identity.

iii) Write the ionic half-equation for the reaction at the anode.

b) The electrolytic purification of silver is similar to the method used to purify copper. Explain how pure silver is obtained from a block of impure silver. Write an ionic half-equation for the reaction that will occur at the anode.

(Edexcel GCE O Level, Paper 2, Jan. 2011, 2(a)–(b))

25 In order to prepare 50 dm^3 of 0.1 M $CuSO_4(aq)$, an inexperienced electroplating worker added the required exact amount of $CuSO_4 \bullet 5H_2O(s)$ to water in a plastic container. He then stirred the mixture with an iron rod until the $CuSO_4 \bullet 5H_2O(s)$ dissolved completely. Finally, he sent a sample of the solution to the Quality Control Laboratory for analysis but found that the concentration of $CuSO_4(aq)$ was lower than 0.1 M .

a) With the aid of a chemical equation, explain why the concentration of the $CuSO_4(aq)$ prepared was lower than 0.1 M.

b) The worker used the prepared $CuSO_4(aq)$ to coat a layer of copper on a metallic object by electrolysis. He used an unreasonably high voltage and found that some bubbles were formed on the object and the copper layer easily flaked off.

i) Explain why copper can be coated on the metallic object by electrolysis.

ii) Suggest what the bubbles were, and explain why the copper layer easily flaked off.

c) Draw a labelled diagram of the experimental set-up used in a laboratory for coating a layer of copper on a metallic object by electrolysis.

(HKDSE, Paper 1B, 2012, 5)

Topic Exercise

Refer to Teacher's Resources Book pages 63–66.

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Multiple choice questions

Refer to Teacher's Resources Book for explanations.

🌶1 Consider the experimental set-ups shown in the figures below:

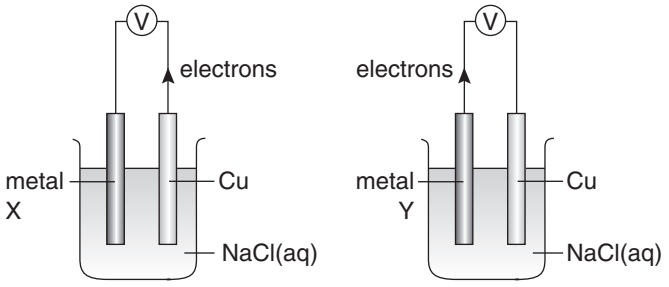


figure 1 figure 2

In figure 1, if the Cu is replaced by Y, which of the following statements is correct?

A Chlorine gas will be evolved at Y.
B A solid will be deposited on the surface of Y.
(C) A greater magnitude of voltage will be recorded.
D Electrons will flow from X to Y through the external circuit.

(HKCEE, Paper 2, 2010, 36)

2 In which of the following species does the underlined element have an oxidation number of +3?

A $\underline{O}_3$
B $H_3\underline{P}O_4$
C $\underline{N}H_4^+$
(D) $\underline{N}O_2^-$

(HKCEE, Paper 2, 2010,11)

3 Consider the following equation:

$$p\underline{Cl}O_3^- + qSO_2 + rH_2O \longrightarrow p\underline{Cl}^- + qSO_4^{2-} + 2rH^+$$

Which of the following combinations is correct?

	p	*q*	*r*
A	1	2	2
(B)	1	3	3
C	2	3	2
D	1	2	3

(HKCEE, Paper 2, 2011, 14)

4 Consider the following chemical equation:

$$2\underline{I}O_3^-(aq) + wH_2O_2(aq) + xH^+(aq) \longrightarrow \underline{I}_2(aq) + yO_2(g) + zH_2O(l)$$

Which of the following is the correct combination of the reaction coefficients *y* and *z*?

	y	*z*
A	4	5
B	5	4
(C)	5	6
D	6	5

(HKDSE, Paper 1A, 2013, 16)

5 Which of the following processes does NOT involve redox reaction?

A Rusting of iron nails
(B) Thermal decomposition of calcium carbonate
C Adding zinc granules to concentrated hydrochloric acid
D Adding magnesium ribbons to copper(II) sulphate solution

(HKCEE, Paper 2, 2011, 6)

6 The tendency of being reduced of six ionic species increases in the order as shown below:

$Ba^{2+}(aq) < Na^{+}(aq) < Mg^{2+}(aq) < H^{+}(aq) < Cu^{2+}(aq) < Hg^{2+}(aq)$

Which of the following statements is correct?

A Ba(s) does NOT react with $H^{+}(aq)$.
B Na(s) has a stronger reducing power than Hg(l).
C $Hg^{2+}(aq)$ is the weakest oxidizing agent among the six species.
D Displacement reaction occurs when Cu(s) is immersed in $MgSO_4(aq)$.

(HKDSE, Paper 1A, 2012, 13)

7 Which of the following statements concerning ammonium iodide is correct?

A Ammonium iodide solution is a weak alkali.
B Ammonium iodide solution is brown in colour.
C Reaction will occur when ammonium iodide is mixed with chlorine water.
D No reaction will occur when ammonium iodide is heated with sodium hydroxide.

(HKCEE, Paper 2, 2009, 31)

8 Which of the following is / are advantage(s) of alkaline manganese cells?

(1) Alkaline manganese cells have a long shelf life.
(2) Alkaline manganese cells are rechargeable.
(3) Alkaline manganese cells of large size produce a higher voltage than smaller ones.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

9 Which of the following statements concerning a zinc-carbon cell are correct?

(1) Manganese(IV) oxide acts as the anode.
(2) Ammonium chloride acts as an electrolyte.
(3) Zinc acts as the negative electrode.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2007, 44)

10 Which of the following substances, when mixed with bromine water, would form a colourless solution?

(1) Sodium sulphite
(2) Sodium chloride
(3) Sodium iodide

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2008, 21)

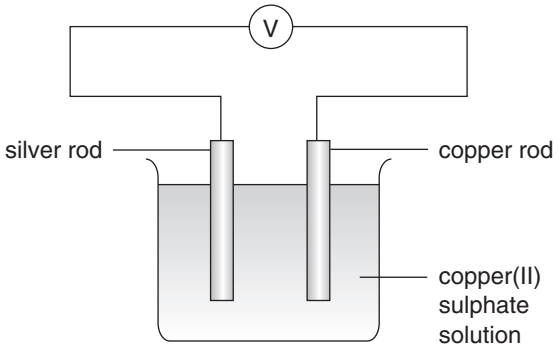
11 Which of the following statements concerning the reaction of concentrated nitric acid with copper is / are correct?

(1) A colourless gas is evolved.
(2) One mole of $NO_3^{-}(aq)$ ions requires one mole of electrons for reduction.
(3) It involves a displacement reaction.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2007, 46)

12 Consider the experimental set-up shown below:



A current has flowed through the external circuit for some time. Which of the following statements is / are correct?

(1) The mass of the cathode increases.
(2) The copper(II) ions are reduced at the copper rod.
(3) The colour of the copper(II) sulphate solution gradually fades out.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

Directions : Each question (Questions 13–15) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :

A	Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B	Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C	The 1st statement is false but the 2nd statement is true.
D	Both statements are false.

	1st statement	2nd statement	
13	Bromine water can react with NaI(aq).	The reducing power of I^-(aq) ion is stronger than that of Br^-(aq) ion.	A
		(HKCEE, Paper 2, 2010, 30)	
14	Sulphates can be oxidized to sulphites.	The oxidation number of sulphur in sulphates is higher than the oxidation number of sulphur in sulphites.	C
		(HKCEE, Paper 2, 2009, 49)	
15	When using a zinc-carbon cell in electrolysis, the carbon electrode of the cell is connected to the cathode of the electrolytic cell.	In a circuit using a zinc-carbon cell to supply electricity, electrons in the external circuit flow to the carbon electrode of the zinc-carbon cell.	C
		(HKCEE, Paper 2, 2010, 49)	

Part II Structured questions

16 In the process of photosynthesis, carbon dioxide and water form glucose and oxygen:

$$6CO_2(g) + 6H_2O(l) \longrightarrow C_6H_{12}O_6(aq) + 6O_2(g)$$

a) What are the oxidation numbers of carbon in CO_2 and $C_6H_{12}O_6$ respectively?

b) Identify the oxidizing agent and reducing agent in the process. Explain your answer in terms of changes in oxidation numbers.

17 During the extraction of bromine industrially, chlorine is bubbled through a solution of bromide ions. A student thought this principle would also work for extracting iodine and carried out the experiment below.

Stage 1 The student bubbled some chlorine through an aqueous solution of potassium iodide.

Stage 2 The student added an organic solvent and shook the mixture.

a) What would the student see at *Stage 1*?

b) Name the products and write an ionic equation for the reaction in *Stage 1*.

c) Why does the reaction in *Stage 1* occur?

d) What would the student see at *Stage 2*?

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jun. 2009, 5(d))

18 This question is about iodine and its compounds.

a) Iodine is formed when an aqueous solution containing iodide ions reacts with an aqueous solution of iron(III) ions.

Write the ionic equation for this reaction.

b) Hydrogen iodide gas is usually prepared by adding phosphoric acid to solid potassium iodide.

Suggest why phosphoric acid is used in this preparation rather than concentrated sulphuric acid.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2011, 13(a)(ii), (b)(i))

19 A bromate of potassium, $KBrO_3$, was widely used for many years to improve the properties of bread dough. However, after baking, the small amounts of $KBrO_3$ remaining in the bread were found to be a health risk.

The problem has been solved by adding the $KBrO_3$ in an aqueous solution with iron(II) sulphate and ascorbic acid. This makes sure that any unwanted $KBrO_3$ is removed on baking.

The bromate ion, BrO_3^-, acts as an oxidizing agent in acidic solution. It reacts with iron(II) ions when the dough is baked.

a) The ionic half-equation for the reaction of BrO_3^- is:

$$BrO_3^-(aq) + 6H^+(aq) + 6e^- \longrightarrow Br^-(aq) + 3H_2O(l)$$

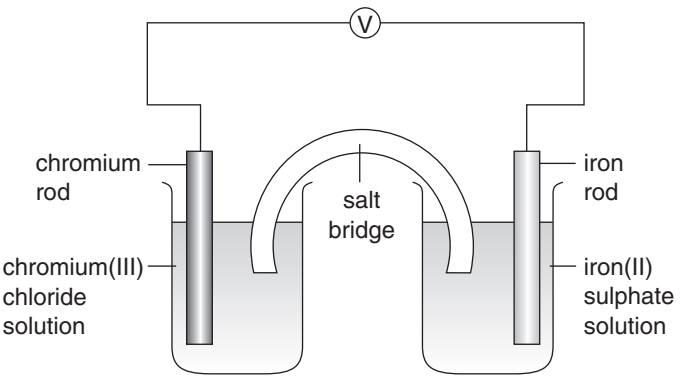
Give the oxidation numbers of Br in this reaction and used them to explain why BrO_3^- is acting as an oxidizing agent.

b) Give a balanced equation for the redox reaction between BrO_3^- and iron(II) ions in acidic solution to produce iron(III) ions.

c) Suggest a reason why ascorbic acid is added to the bread dough.

(OCR Advanced GCE, Chem. B (Salters), F334, Jan. 2012, 4(a))

20 The diagram below shows the set-up of a simple chemical cell. As time goes by, the colour of iron(II) sulphate solution in the beaker gradually fades out.



Topic Exercise

a) State, with explanation, the direction of electron flow in the external circuit.

b) Write an ionic half-equation for the change occurring at the chromium rod.

c) i) Chromium is one of the components of stainless steel. Suggest how chromium can prevent iron in stainless steel from rusting.

ii) Coating chromium on iron-made objects can prevent the objects from rusting. Name this coating process and explain how this process can prevent rusting.

(HKCEE, Paper 1, 2011, 4)

21 Nickel metal hydride cells (NiMH cells) are being developed for possible use in cars. In a NiMH cell, an alloy is used to absorb hydrogen as metal hydride. For simplicity, the alloy can be represented as M and the metal hydride as MH.

The overall cell reaction in a NiMH cell is shown below.

$$MH + NiO(OH) \longrightarrow M + Ni(OH)_2$$

The ionic half-equation at one electrode is shown below.

$$NiO(OH) + H_2O + e^- \longrightarrow Ni(OH)_2 + OH^-$$

a) Deduce the ionic half-equation at the other electrode.

b) State a method, other than absorption, that is being developed to store hydrogen for possible use as a fuel in cars.

(OCR Advanced GCE, Chem. A, F325, Jan. 2012, 5(c))

22 A type of breathalyzer for investigating drink-driving consists of a chemical cell. The breath of the driver is allowed to get into contact with one of the electrodes of the cell. If the breath contains ethanol, the ethanol would be converted to ethanoic acid at this electrode and an electric current would be produced.

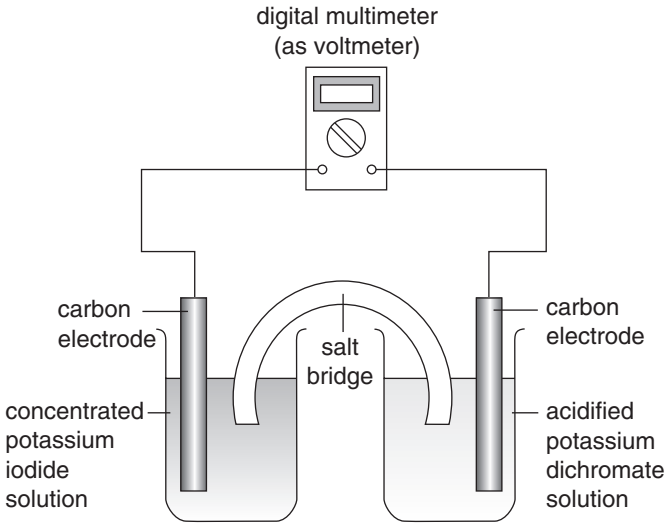
a) Explain whether the above-mentioned electrode acts as the anode or cathode of the chemical cell.

b) Write an ionic half-equation for the change occurring at this electrode.

c) Explain how this type of breathalyzer could estimate the amount of ethanol in the breath of the driver.

(HKCEE, Paper 1, 2011, 10(b))

23 Consider the chemical cell shown below:



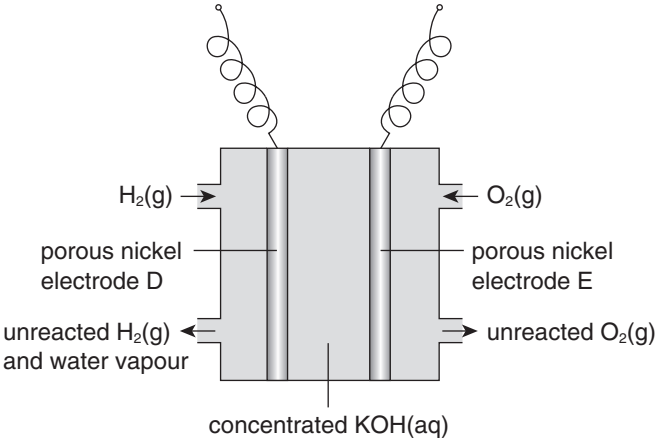
a) Suggest the observable change for the acidified potassium dichromate solution after some time. Write an ionic half-equation for the reaction that occurs.

b) i) Suggest the observable change for the potassium iodide solution after some time. Write an ionic half-equation for the reaction that occurs.

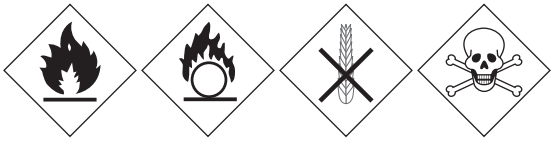
ii) Explain whether the iodide ions are oxidized or reduced.

c) Identify the direction of electron flow in the external circuit.

24 The diagram below shows the structure of a hydrogen-oxygen fuel cell using concentrated potassium hydroxide solution as the electrolyte.



a) An oxygen cylinder can be used to provide oxygen for the above fuel cell. From the hazard warning labels shown below, circle the label that should be displayed on the oxygen cylinder.



b) Write the ionic half-equation for the change occurring at each of the following electrodes when this fuel cell is producing a current.

Electrode D

Electrode E

c) Some people have the view that cars powered by hydrogen-oxygen fuel cells are more environmentally friendly than those powered by petrol.

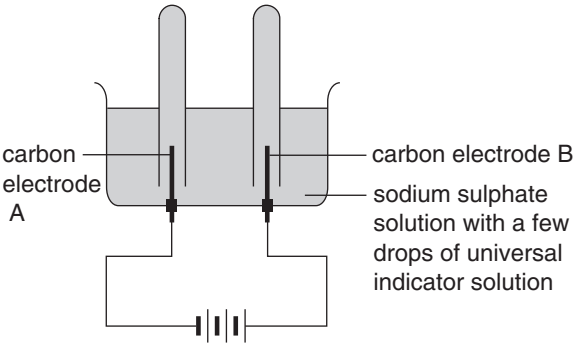
Comment on this view from each of the following aspects:

i) source of fuel

ii) the car emissions

(HKDSE, Paper 1B, 2013, 10)

25 The diagram below shows the set-up used in an investigation on the electrolysis of dilute sodium sulphate solution.

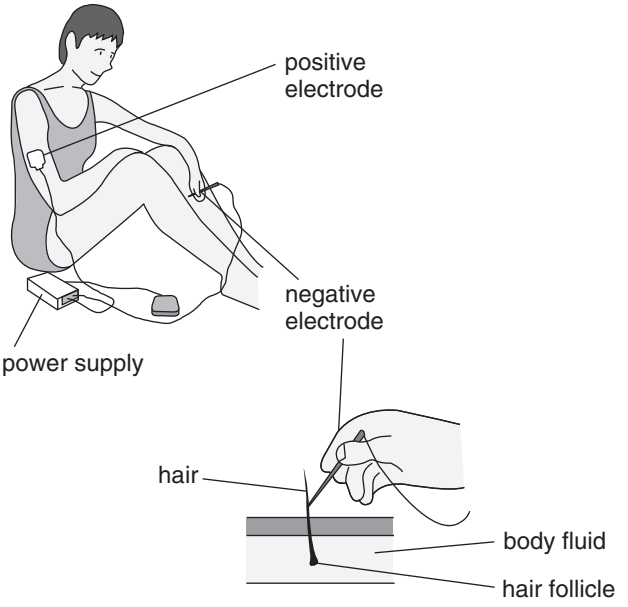


a) Write ionic half-equations for the reaction that occur at electrodes A and B respectively.

b) What will be the colour of the solution near to each electrode after some time? Explain.

c) What would be the change in the electrolyte after electrolysis? Explain.

26 Electrolysis can be used to remove unwanted hair from the skin.



The hair is first coated with a layer of gel containing ions in solution.

The positive electrode is connected by a patch to the skin.

The negative electrode is connected to the hair. Electricity flows through the gel and causes electrolysis of the body fluid around the hair follicle.

a) Explain why the gel containing ions in solution can conduct electricity.

b) The body fluid is a solution that contains sodium chloride. The electricity causes the electrolysis of a small amount of this solution.

This solution contains hydrogen ions that move to the negative electrode.

i) The ionic half-equation represents the reaction at the negative electrode.

$2H^+ + 2e^- \longrightarrow H_2$

Explain why this reaction is a reduction.

ii) As a result of the electrolysis of sodium chloride solution, an alkali forms and kills the hair follicle.

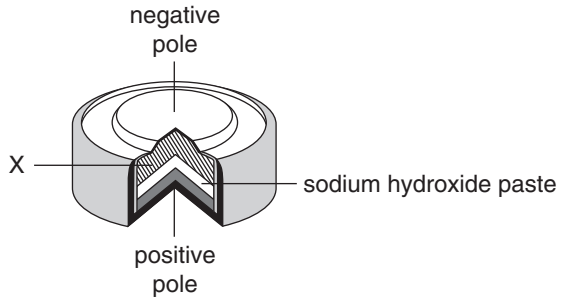
What is the name of this alkali?

iii) Complete the ionic half-equation for the reaction at the positive electrode.

$Cl^- \longrightarrow \quad Cl_2$

(AQA GCSE (Higher Tier), Additional Science, Unit Chemistry C2, Jan. 2009, 7(b)–(c))

27 The diagram below shows a kind of traditional 'button cell' making from mercury(II) oxide, zinc powder and sodium hydroxide paste:



When the cell is producing a current, the overall cell reaction can be represented by the following chemical equation:

$Zn + HgO \longrightarrow ZnO + Hg$

a) Explain whether mercury(II) oxide or zinc powder should be at the region labelled X.

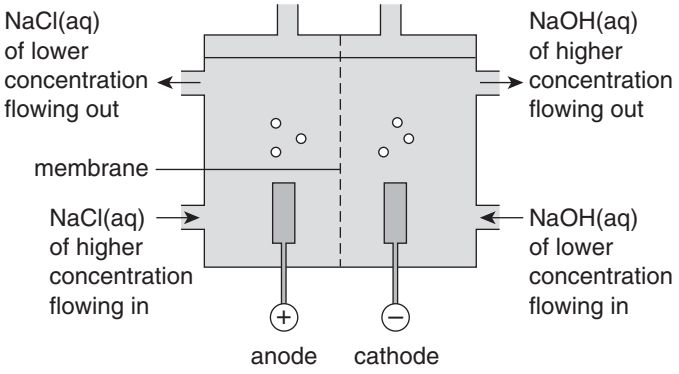
b) What is the function of the sodium hydroxide paste in the cell?

c) Why should this kind of button cell be banned in the market?

d) Explain whether the cell can work if mercury(II) oxide is replaced by magnesium oxide, while other materials remain unchanged.

e) Explain the change of the maximum voltage supplied by the cell if zinc powder is replaced by copper powder, while other materials remain unchanged.

f) The following diagram shows the electrolytic cell used in the manufacture of sodium hydroxide:



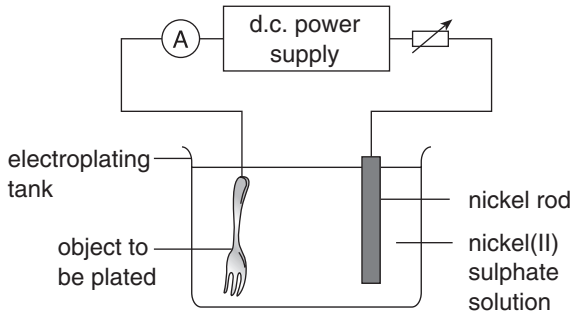
i) Write an ionic half-equation for the anodic reaction.

ii) Write an ionic half-equation for the cathodic reaction.

iii) It is known that only cations can pass through the membrane. Explain why sodium hydroxide solution of higher concentration is eventually obtained.

(HKCEE, Paper 1, 2010, 11)

28 The diagram below shows the set-up for electroplating an object with nickel.



a) Explain the term 'electroplating'.

b) Explain why nickel(II) sulphate solution can conduct electricity.

c) Which is the anode, the nickel electrode or the object to be plated?

d) Write an ionic half-equation for the reaction taking place at the object to be plated.

e) Explain why it is better to use a nickel electrode than a carbon electrode in the above process.

f) In a nickel-plating factory, the wastewater is treated to remove nickel(II) ions before discharge.

i) Suggest TWO reasons why it is necessary to remove nickel(II) ions from the wastewater before discharge.

ii) Suggest how nickel(II) ions can be removed from the wastewater.

29 The method of extraction of zinc has changed as different ores containing the element have been discovered and as technology has improved.

Extraction Process 1

In the earliest process, calamine (impure zinc carbonate) was heated with charcoal in earthenware pots. This two-stage process gave a low yield of zinc.

$ZnCO_3(s) \longrightarrow ZnO(s) + CO_2(g)$

$ZnO(s) + C(s) \longrightarrow Zn(s) + CO(g)$

Extraction Process 2

Deposits of calamine were being used up and a new two-stage process was developed using zinc sulphide ores. All of the waste gases from this process were released into the atmosphere.

$2ZnS(s) + 3O_2(g) \longrightarrow 2ZnO(s) + 2SO_2(g)$

$ZnO(s) + C(s) \longrightarrow Zn(s) + CO(g)$

Extraction Process 3

The modern process uses the electrolysis of aqueous solutions of very pure zinc sulphate. The first step in this process is the same as the first step in *Extraction Process 2*. The second step uses sulphuric acid made from the SO_2 collected in the first step. The third step involves the electrolysis of zinc sulphate solution to form pure zinc.

$2ZnS(s) + 3O_2(g) \longrightarrow 2ZnO(s) + 2SO_2(g)$

$ZnO(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2O(l)$

$ZnSO_4(aq) \xrightarrow{\text{electrolysis}} Zn(s)$

a) State and explain one environmental reason why *Extraction Process 3* is an improvement over *Extraction Process 2*.

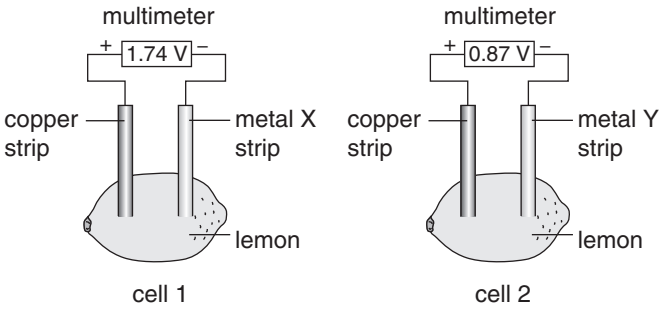
b) Give one reason why *Extraction Process 3* is an expensive method of making zinc but one which is justified in terms of the product formed.

c) Deduce the ionic half-equation for the formation of zinc from zinc ions during the electrolysis of zinc sulphate solution and identify the electrode at which this reaction occurs.

d) Identify one reaction from the three extraction processes that is NOT a redox reaction and state the type of reaction that it is. In terms of redox, state what happens to the carbon in *Extraction Process 2*.

(AQA Advanced Subsidiary GCE, Unit 2, Jun. 2010, 9(b)–(e))

30 Consider the information concerning the lemon cells shown in the diagrams below:



a) What is the function of the lemons in these cells?

b) By completing the table below, arrange metal X, metal Y and copper in increasing order of reducing power.

Reducing power increasing →		

c) For cell 1, write the ionic half-equation for the change that occurs at:

i) the metal X strip (X is a group II metal), and

ii) the copper strip.

d) For cell 2, would the metal Y strip be the positive electrode if the copper strip is replaced with a silver strip? Explain your answer.

(HKDSE, Paper 1B, 2012, 3)